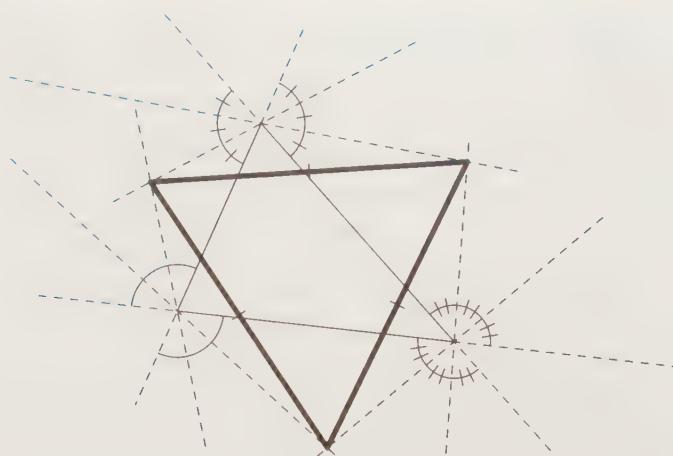




Morley's Second Triangle

New facts often trigger new ideas. -- Alex F. Osborn

17

CHAPTER

Analytic Geometry

Much of the mathematics developed by the ancient Greeks was geometry. In fact, even much of the number theory and algebra that they developed was expressed in geometric terms. However, by the 16th or 17th century A.D., algebra was recognized as an extremely important field of mathematics in its own right, possibly equaling, or even exceeding, geometry.

In 1637, the great mathematician and philosopher René Descartes brought these two great fields of mathematics together when he described a general method to represent geometric figures with algebraic equations. This combination of algebra and geometry is often referred to as **analytic geometry**. In this chapter, we review how to describe lines and circles with equations, then we apply these methods to both algebra and geometry problems.

WARNING!!



If you have not studied graphing lines and circles in the Cartesian plane, you may find the material in this chapter extremely difficult. If so, we recommend studying the fundamentals of graphing lines and circles in our *Introduction to Algebra* textbook.

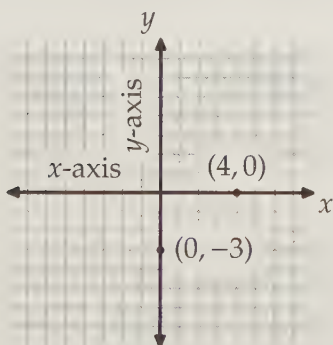
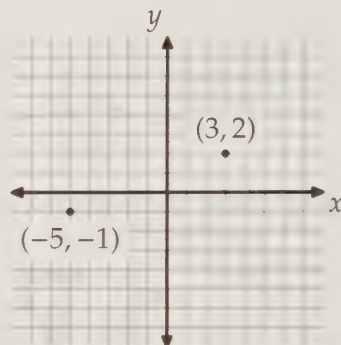
Moreover, the algebraic manipulation required for parts of this chapter is considerably more complicated than in the rest of this text. Specifically, if you are not comfortable with completing the square, you may wish to study it in *Introduction to Algebra* (or another text) before returning to this chapter.

Descartes's great insight that unified geometry and algebra has become so important that we still use Descartes's name to describe the result. We use the **Cartesian plane** to describe geometric figures with algebraic equations.

At right is the Cartesian plane. The center of the plane, where the bold lines meet, is called the **origin**. On the Cartesian plane each point is represented by an **ordered pair** of numbers. These numbers denote the position of the point *relative to the origin*.

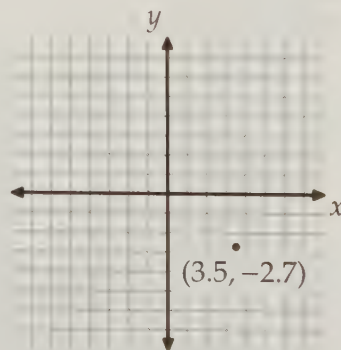
For example, we denote the point that is 3 steps to the right and 2 steps up from the origin with the ordered pair $(3, 2)$. The "ordered" part is very important! The first number always denotes how far the point is to the right (or left) of the origin, and the second number denotes how far the point is above (or below) the origin. We call the two numbers in an ordered pair the **coordinates** of the point. By convention, we call the horizontal (left-right) coordinate the **x -coordinate** and we call the vertical (up-down) coordinate the **y -coordinate**.

As you might have guessed, the x -coordinate of a point is negative when the point is to the left of the origin, and the y -coordinate is negative when the point is below the origin. For example, the point $(-5, -1)$ shown in the diagram is 5 steps to the left and one step below the origin. The point $(0, 0)$ is 0 steps to the right and 0 steps up from the origin. In other words, $(0, 0)$ is the origin itself.



When the point is neither above nor below the origin, its y -coordinate is 0. Such a point is directly to the left or right (or on) the origin. Therefore, the point must be on the bold horizontal line in the diagram at left. We call this line the **x -axis**. Similarly, the vertical line consisting of points directly above or below (or on) the origin is called the **y -axis**. We usually label the x -axis and y -axis with an x and a y , respectively, as shown in each of our diagrams.

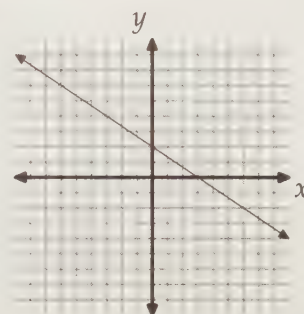
Finally, we aren't restricted to using integers for our coordinates. We can plot any point represented by an ordered pair of two real numbers on the Cartesian plane. For example, the diagram at right depicts the point $(3.5, -2.7)$, which is 3.5 steps to the right and 2.7 steps below the origin. Points that have integers for both coordinates are called **lattice points**.



17.1 Lines

In this section, we review how equations can be used to represent lines. We cover this material in much more detail in *Introduction to Algebra*. If you are unfamiliar with using equations to represent lines, you may wish to study the fundamentals covered in *Introduction to Algebra* before continuing.

The **graph** of an equation that has x and y as variables consists of all points (x, y) on the Cartesian plane that satisfy the equation. For example, the graph of the equation $2x + 3y = 6$ is shown at right. As we see, the graph is a line. Indeed, the graph of any equation of the form $Ax + By = C$, where A , B , and C are constants and A and B are not both 0, is a line. As a result, such an equation is often called a **linear equation**.



The points where a graph intersects the x -axis are called the **x -intercepts** of the graph, and the points where a graph intersects the y -axis are called the **y -intercepts** of the graph. For example, the only x -intercept of the graph of $2x + 3y = 6$ at right is $(3, 0)$ and the only y -intercept of the graph is $(0, 2)$.

One way we describe graphed lines is by indicating the 'direction' of the line, which we call the **slope** of the line. We define the slope, m , of the line that passes through the points (x_1, y_1) and (x_2, y_2) to be

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

A line with positive slope goes upward as it goes from left to right, and a line with negative slope goes downward as it goes from left to right.

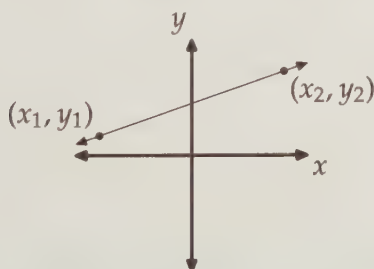


Figure 17.1: A Line With Positive Slope

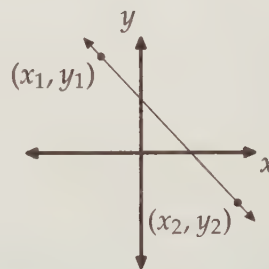
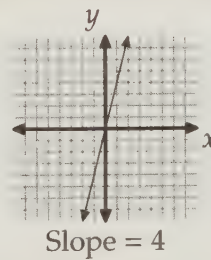
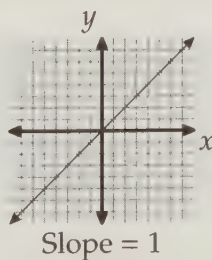
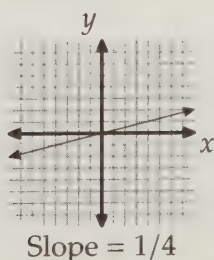


Figure 17.2: A Line With Negative Slope

The slope of a horizontal line is equal to 0, and the slope of a vertical line is undefined. Moreover, the greater the magnitude of the slope of a line, the more 'steep' the line is.



Problem

Problem 17.1: Find an equation whose graph is the line passing through the points $(-1, 7)$ and $(-5, -3)$.

Problem 17.2: Find an equation whose graph is the line with slope 4 and y -intercept $(0, -3)$.

Problem 17.3:

- Find the length of the segment with endpoints $(-4, -3)$ and $(6, 1)$.
- Find a formula for the distance between the points (x_1, y_1) and (x_2, y_2) .

Problem 17.4: Let point A be (x_1, y_1) and point B be (x_2, y_2) . Let M be the point with coordinates

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right).$$

In this problem we show that M is the midpoint of $\overline{AB}$.

- Show that $AM = MB$.
- Show that M is on $\overleftrightarrow{AB}$.

Problem 17.5: Explain why two different lines that have the same slope are parallel.

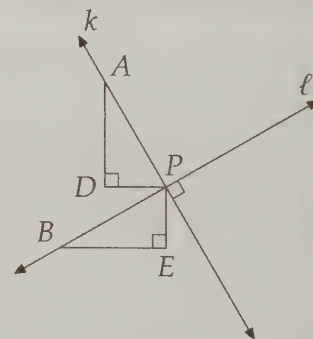
Problem 17.6: In this problem we show that if two lines are perpendicular, and neither line is vertical, then the product of their slopes is -1 .

Shown at right are perpendicular lines k and ℓ . Let P be the point where these lines intersect. We construct two right triangles such that their hypotenuses are along lines k and ℓ , respectively, as shown. Moreover, each right triangle has a horizontal leg and a vertical leg.

- Show that $\triangle ADP \sim \triangle BEP$.
- Use the similar triangles from part (a) to show that

$$\frac{DA}{DP} \cdot \frac{EP}{EB} = 1.$$

Why does this tell us that the products of the slopes of k and ℓ is -1 ?



We start with a review of some different forms in which we can write equations whose graphs are lines.

Problem 17.1: Find an equation whose graph is the line passing through the points $(-1, 7)$ and $(-5, -3)$.

Solution for Problem 17.1: The slope of the line through these two points is

$$\frac{7 - (-3)}{-1 - (-5)} = \frac{10}{4} = \frac{5}{2}.$$

Therefore, if (x, y) is any point on this line other than $(-1, 7)$, then the slope between (x, y) and $(-1, 7)$ must be $5/2$. So, we must have

$$\frac{y - 7}{x - (-1)} = \frac{5}{2}.$$

Multiplying both sides of this equation by $x - (-1)$ gives us $y - 7 = \frac{5}{2}[x - (-1)]$. This is an example of the **point-slope form** of a linear equation.

Important: The graph of the equation



$$y - y_1 = m(x - x_1)$$

is a line through (x_1, y_1) with slope m . This is called a **point-slope form** of the equation.

While we often use point-slope form to find an equation of a line given the slope of the line and a point on the line, we often then convert the equation to **standard form**.

Important: The **standard form** of a linear equation is $Ax + By = C$, where, if possible, A , B , and C are integers, A is positive, and A , B , and C have no common factors besides 1.



We start converting $y - 7 = \frac{5}{2}[x - (-1)]$ to standard form by first multiplying both sides by 2 to get $2(y - 7) = 5(x + 1)$. Expanding both sides and rearranging gives $5x - 2y = -19$.

We can check our answer by making sure that the two points $(x, y) = (-1, 7)$ and $(x, y) = (-5, -3)$ satisfy our equation. Both do, so we know the equation $5x - 2y = -19$ represents the line through $(-1, 7)$ and $(-5, -3)$. $\square$

One more form that is occasionally useful is **slope-intercept form**, which we'll investigate in the next problem.

Problem 17.2: Find an equation whose graph is the line with slope 4 and y -intercept $(0, -3)$.

Solution for Problem 17.2: Because the line passes through $(0, -3)$ and has slope 4, a point-slope form of the line is $y - (-3) = 4(x - 0)$. Isolating y on the left side then gives us $y = 4x - 3$. Notice that the coefficient of x equals the slope of our line and the constant term on the right equals the y -coordinate of the y -intercept. This is the slope-intercept form of the equation:

Important: The **slope-intercept form** of a linear equation is



$$y = mx + b,$$

where m is the slope of the line and b is the y -coordinate of the y -intercept.

As an Exercise, you'll prove that the graph of an equation of the form $y = mx + b$ has slope m . $\square$

We'll now review some particularly useful relationships regarding segments and lines on the Cartesian plane.

Problem 17.3:

- Find the length of the segment with endpoints $(-4, -3)$ and $(6, 1)$.
- Find a formula for the distance between the points (x_1, y_1) and (x_2, y_2) .

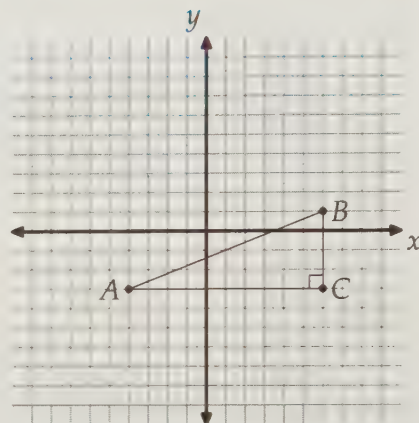
Solution for Problem 17.3:

- We start by labeling our points A and B such that A is $(-4, -3)$ and B is $(6, 1)$.

One of our favorite tactics for finding the length of a segment in a geometry problem is to build a right triangle with that segment as a side. Here, an easy way to build a right triangle with $\overline{AB}$ as a side is to extend segments horizontally from A and vertically from B to meet at point C , as shown. Since a horizontal line and a vertical line meet at a right angle, $\angle ACB$ is a right angle. So, if we can find AC and BC , we can use the Pythagorean Theorem to find AB .

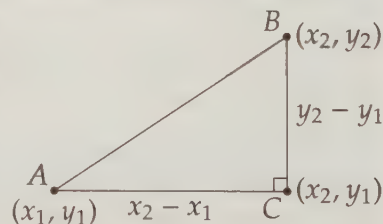
The x -coordinate of C is the same as the x -coordinate of B and the y -coordinate of C is the same as the y -coordinate of A . So, point C is $(6, -3)$. Therefore, $AC = 6 - (-4) = 10$ and $BC = 1 - (-3) = 4$, and the Pythagorean Theorem gives us

$$AB = \sqrt{AC^2 + BC^2} = \sqrt{100 + 16} = \sqrt{116} = 2\sqrt{29}.$$



- Building a right triangle worked well when finding the distance between two specific points, so let's try using that tactic to find a formula for the distance between (x_1, y_1) and (x_2, y_2) . Using part (a) as a guide, we let A be (x_1, y_1) and B be (x_2, y_2) and build a right triangle with $\overline{AB}$ as a side.

In the diagram at right, we assume that B is above and to the right of A . As before, we draw a vertical line down from B and a horizontal line to the right from A . We label the point where these lines meet C . Since C is directly below B , its x -coordinate is the same as B 's, or x_2 (in other words, it is just as far horizontally from the y -axis as B is). Similarly, since C is directly to the right of A , its y -coordinate is the same as A 's, y_1 .



Now we're ready to use the Pythagorean Theorem. We can use our coordinates to see that C is $x_2 - x_1$ to the right of A and $y_2 - y_1$ below B . So, we can use the Pythagorean Theorem to find

$$AB^2 = AC^2 + BC^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2.$$

We can take the square root of both sides of this equation to find a formula for the distance between the points (x_1, y_1) and (x_2, y_2) :

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

This is often called the **distance formula**. Notice that if $x_1 = x_2$, then the distance between (x_1, y_1) and (x_2, y_2) is just the nonnegative difference between the y -coordinates. Let's see if that's what

the distance formula gives us when $x_1 = x_2$:

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{(x_2 - x_2)^2 + (y_2 - y_1)^2} = \sqrt{(y_2 - y_1)^2} = |y_2 - y_1|.$$

Indeed, the distance formula gives the expected result when $x_1 = x_2$. (We use absolute value signs when writing $\sqrt{(y_2 - y_1)^2} = |y_2 - y_1|$ because square roots, like distances, must be nonnegative.) Similarly, the distance formula tells us that the distance between (x_1, y_1) and (x_2, y_2) when $y_1 = y_2$ is the nonnegative difference between the x -coordinates.

□

Important: The distance in the plane between the points (x_1, y_1) and (x_2, y_2) is



$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

This is often referred to as the **distance formula**.

The distance formula is essentially the same thing as the Pythagorean Theorem. In fact, because you know the Pythagorean Theorem, you basically know the distance formula already, so you shouldn't have to memorize the distance formula.

Problem 17.4: Let point A be (x_1, y_1) and point B be (x_2, y_2) . Let M be the point with coordinates

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right).$$

Show that M is the midpoint of $\overline{AB}$.

Solution for Problem 17.4: In order to show that M is the midpoint of $\overline{AB}$, we must show that $AM = MB$ and that M is on $\overleftrightarrow{AB}$. To show that $AM = MB$, we turn to the distance formula. We have

$$AM = \sqrt{\left(\frac{x_1 + x_2}{2} - x_1 \right)^2 + \left(\frac{y_1 + y_2}{2} - y_1 \right)^2} = \sqrt{\left(\frac{x_2 - x_1}{2} \right)^2 + \left(\frac{y_2 - y_1}{2} \right)^2},$$

$$MB = \sqrt{\left(x_2 - \frac{x_1 + x_2}{2} \right)^2 + \left(y_2 - \frac{y_1 + y_2}{2} \right)^2} = \sqrt{\left(\frac{x_2 - x_1}{2} \right)^2 + \left(\frac{y_2 - y_1}{2} \right)^2}.$$

So, we have $AM = MB$.

To show that M is on $\overleftrightarrow{AB}$, we compare the slopes of $\overleftrightarrow{AM}$ and $\overleftrightarrow{MB}$. We have

$$\text{Slope of } \overleftrightarrow{AM} = \frac{\frac{y_1 + y_2}{2} - y_1}{\frac{x_1 + x_2}{2} - x_1} = \frac{y_2 - y_1}{x_2 - x_1},$$

$$\text{Slope of } \overleftrightarrow{MB} = \frac{y_2 - \frac{y_1 + y_2}{2}}{x_2 - \frac{x_1 + x_2}{2}} = \frac{y_2 - y_1}{x_2 - x_1}.$$

When $x_2 \neq x_1$, the slopes of $\overleftrightarrow{AM}$ and $\overleftrightarrow{MB}$ are the same, so A , M , and B are collinear. If $x_2 = x_1$, then A , M and B are on the same vertical line.

Since A , M , and B are collinear and $AM = MB$, we know that M is the midpoint of $\overline{AB}$. □

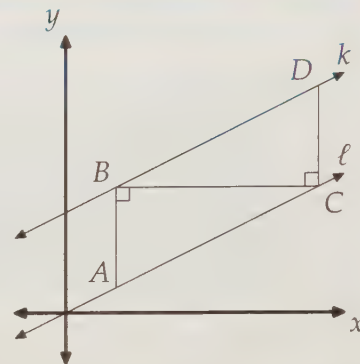
Important: The midpoint of the segment with endpoints (x_1, y_1) and (x_2, y_2) is



$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right).$$

Problem 17.5: Explain why two different lines with the same slope are parallel.

Solution for Problem 17.5: Slope is a measure of the direction of a line. Therefore, we expect two lines with the same slope to be parallel. We can prove this quickly with geometry. If our two lines both have slope 0, then they are both horizontal, and clearly are parallel. Otherwise, we can draw a horizontal line segment with one endpoint on each line, as shown in the diagram at right. We can build two right triangles by drawing two vertical line segments, as shown. If we show that $\angle ACB = \angle CBD$, then we can deduce that $k \parallel \ell$.



The only given information we have is that the lines have the same slope, so we need to find some way to express the slope of each line to allow us to use this information. Fortunately, our diagram gives us the answer. The slope of k is DC/BC (change in y -coordinate divided by change in x -coordinate), and the slope of line ℓ is AB/BC . Because the slopes of k and ℓ are equal, we have $DC/BC = AB/BC$, so $DC = AB$. We also have $\angle ABC = \angle BCD$, so we have $\triangle ABC \cong \triangle DCB$ by SAS Congruence. Therefore, we have $\angle ACB = \angle CBD$, so $k \parallel \ell$, as desired.

You'll have the chance to prove the converse, that two parallel lines have the same slope (if they are not vertical) as an Exercise. $\square$

Important: If two lines have the same slope, then they are parallel. Conversely, if two non-vertical lines are parallel, then they have the same slope.



We've seen that the relationship between the slopes of two parallel lines is pretty straightforward. What about perpendicular lines?

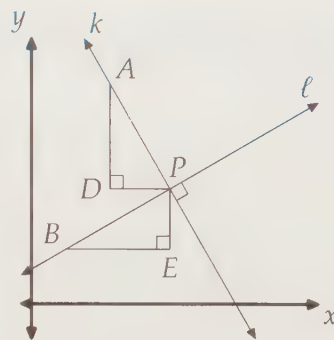
Problem 17.6: Show that if two lines are perpendicular, and neither line is vertical, then the product of their slopes is -1 .

Solution for Problem 17.6: Because the lines are perpendicular and neither is vertical, one line has positive slope and the other has negative slope. To see why, suppose the lines intersect at P , and that their slopes have the same sign. Then, the smaller angle formed by the lines is smaller than the right angle formed by the vertical line and horizontal line through P . So, if the slopes of the lines have the same sign, the lines cannot be perpendicular.

Let the line with negative slope be k and the line with positive slope be ℓ . Building right triangles worked so well in the last problem that we should try it again for this problem. We're also guided to build right triangles since these will give us a way to express the slopes of k and ℓ . We can use essentially this same proof for any configuration of k and ℓ in which the lines are perpendicular and neither line is vertical.

We make right triangles $\triangle ADP$ and $\triangle PBE$ such that each has a horizontal leg and a vertical leg. From $\triangle PBE$, we see that the slope of ℓ is EP/EB . From $\triangle ADP$, we see that the slope of k is $-DA/DP$. Make sure you see why we need the negative sign; the slope of k is clearly negative. Now that we have expressions for the slopes of k and ℓ , we can write an equation we wish to prove. We must show that

$$\left(\frac{EP}{EB}\right)\left(-\frac{DA}{DP}\right) = -1.$$



The ratios in this expression make us think of similar triangles. The right triangles are a pretty big clue, too. We already have $\angle ADP = \angle BEP$, so we only need to find one more pair of equal angles to prove the right triangles are similar. We probably need to use the fact that k and ℓ are perpendicular, so we focus on that. Since $\angle APB = \angle DPE = 90^\circ$, we have

$$\angle APD = 90^\circ - \angle DPB = \angle BPE.$$

Combining $\angle APD = \angle BPE$ with $\angle ADP = \angle BEP$ gives us $\triangle ADP \sim \triangle BEP$ by AA Similarity. This similarity gives us

$$\frac{DA}{DP} = \frac{EB}{EP}.$$

Multiplying both sides by -1 and by EP/EB gives us the desired $(EP/EB)(-DA/DP) = -1$. Therefore, the product of the slopes of k and ℓ is -1 .

As an Exercise, you'll have the chance to prove the converse: if the product of the slopes of two lines is -1 , then the two lines are perpendicular. $\square$

Important:  If two non-vertical lines are perpendicular, then the product of their slopes is -1 . Conversely, if the product of the slopes of two lines is -1 , then the two lines are perpendicular.

Exercises

- 17.1.1** Find an equation, in standard form, whose graph is a line through $(-2, 5)$ and $(-5, 9)$.
- 17.1.2** Find the x -intercept and y -intercept of the line with slope 4 that passes through $(-7, 2)$.
- 17.1.3** The distance between $(-4, 3)$ and $(a, 0)$ is 5. Find all possible values of a .
- 17.1.4** When finding the distance between two points using the distance formula, does it matter which point we call (x_1, y_1) and which we call (x_2, y_2) ? Why or why not?
- 17.1.5** Point P is $(-5, 2)$ and point Q is $(-8, 8)$.
- Find PQ .
 - Find the midpoint of $\overline{PQ}$.
 - Find T such that Q is the midpoint of $\overline{PT}$.

17.1.6 Line k is perpendicular to the graph of $4x - 3y + 14 = 0$ and passes through $(-3, 2)$. Find an equation in standard form whose graph is line k .

17.1.7 Show that the graph of $y = mx + b$, where m and b are constants, has slope m .

17.1.8 Show that if the product of the slopes of lines k and ℓ is -1 , then $k \perp \ell$.

17.1.9 Suppose A , B , and C are constants and $B \neq 0$. Show that the slope of the graph of the equation $Ax + By + C = 0$ is $-A/B$.

17.2 Circles

In this section, we use the distance formula to find equations whose graphs are circles.

Problems

Problem 17.7:

- Find an equation whose graph is a circle with center $(4, -5)$ and radius $3\sqrt{2}$.
- Explain why the graph of the equation

$$(x - h)^2 + (y - k)^2 = r^2,$$

where h , k , and r are constants with $r > 0$, is a circle with center (h, k) and radius r .

Problem 17.8: Find the area of the region that is enclosed by the graph of the equation

$$2x^2 - 8x + 2y^2 + 16y = 4.$$

Problem 17.7:

- Find an equation whose graph is a circle with center $(4, -5)$ and radius $3\sqrt{2}$.
- Explain why the graph of the equation

$$(x - h)^2 + (y - k)^2 = r^2,$$

where h , k , and r are constants with $r > 0$, is a circle with center (h, k) and radius r .

Solution for Problem 17.7:

- If a point (x, y) is on the circle, then it must be $3\sqrt{2}$ from the center of the circle, which is $(4, -5)$. So, the distance between (x, y) and $(4, -5)$ is $3\sqrt{2}$. The distance formula then gives us

$$\sqrt{(x - 4)^2 + (y + 5)^2} = 3\sqrt{2}.$$

Squaring both sides of this equation gives

$$(x - 4)^2 + (y + 5)^2 = 18.$$

We could multiply out both $(x - 4)^2$ and $(y + 5)^2$, but leaving the equation in this form makes it particularly easy to see that the graph of this equation is a circle with center $(4, -5)$ and radius $3\sqrt{2}$. In our next part, we'll see why.

(b) Taking the square root of both sides of $(x - h)^2 + (y - k)^2 = r^2$ gives us

$$\sqrt{(x - h)^2 + (y - k)^2} = r.$$

Since h and k are constants and r is a positive constant, this equation tells us that the point (x, y) is a distance of r from the point (h, k) . Therefore, every point (x, y) that satisfies this equation is on the circle with center (h, k) and radius r .

Conversely, if a point (x, y) is a distance of r from the point (h, k) , then the distance formula tells us that

$$\sqrt{(x - h)^2 + (y - k)^2} = r.$$

Squaring both sides of this equation gives us $(x - h)^2 + (y - k)^2 = r^2$, so we see that every point on the circle with center (h, k) and radius r satisfies the given equation. Therefore, the graph of the equation $(x - h)^2 + (y - k)^2 = r^2$ is a circle with center (h, k) and radius r .

□

Important: The **standard form** of an equation whose graph is a circle is



$$(x - h)^2 + (y - k)^2 = r^2,$$

where h, k , and r are constants with $r > 0$. The center of the circle is (h, k) and the radius of the circle is r .

Sometimes we have to do a little work to put an equation in standard form.

Problem 17.8: Find the area of the region that is enclosed by the graph of the equation

$$2x^2 - 8x + 2y^2 + 16y = 4.$$

Solution for Problem 17.8: The $2x^2$ and $2y^2$ terms make us think of the standard form of a circle, because expanding the squares on the left side of

$$(x - h)^2 + (y - k)^2 = r^2$$

will give us an x^2 and y^2 . So, we try to write the given equation in this form. Our first step is to divide both sides by 2, to make the coefficients of x^2 and y^2 both equal to 1. This gives us

$$x^2 - 4x + y^2 + 8y = 2.$$

We then complete the square in both x and y by adding 4 and 16 to both sides:

$$x^2 - 4x + 4 + y^2 + 8y + 16 = 2 + 4 + 16.$$

Since $x^2 - 4x + 4 = (x - 2)^2$ and $y^2 + 8y + 16 = (y + 4)^2$, we have

$$(x - 2)^2 + (y + 4)^2 = 22.$$

This equation is in the desired standard form of a circle. Therefore, we know that the graph of this equation is a circle with center $(2, -4)$ and radius $\sqrt{22}$.

WARNING!!

Make sure you see why the center is not $(-2, 4)$ or $(2, 4)$. The standard form of a circle is

$$(x - h)^2 + (y - k)^2 = r^2.$$

Comparing this to

$$(x - 2)^2 + (y + 4)^2 = 22$$

gives us $h = 2$ and $k = -4$.

Because the graph is a circle with radius $\sqrt{22}$, its area is $(\sqrt{22})^2\pi = 22\pi$. $\square$

Exercises

17.2.1 Find the center and the radius of the circle that is the graph of the equation $3x^2 - 12x + 3y^2 + 6y = 15$.

17.2.2 Find the standard form of the equation whose graph is a circle with center $(-2, 7)$ that passes through $(-5, 9)$.

17.2.3 What is the length of the longest chord of the graph of

$$x^2 + 2x + y^2 - 6y = 6$$

that passes through the point $(-2, 4.5)$?

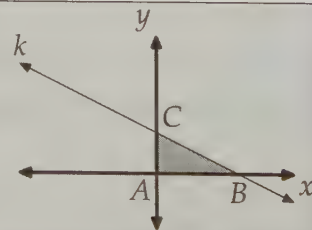
17.2.4★ Find the two points at which the graphs of $2x - y = 7$ and $x^2 - 10x + y^2 + 4y = -4$ intersect.

17.3 Basic Analytic Geometry Problems

In this section, we begin to explore the power of analytic geometry. Because analytic geometry gives us a tool to relate algebra and geometry, it gives us a way to apply geometric tools to algebraic problems, and to use algebraic tools to solve geometry problems.

Problems

Problem 17.9: The area of the shaded region between line k and the axes in the diagram at right is 36. If line k passes through $(12, 0)$, then what is the slope of k ?



Problem 17.10: Let A be $(-5, 6)$, B be $(-7, 9)$, and C be $(1, 10)$.

- Find the length of the median of $\triangle ABC$ from B to $\overline{AC}$.
- Find the slope of each side of $\triangle ABC$. Notice anything interesting?
- Find the area of $\triangle ABC$.
- Find the length of the altitude from A to $\overline{BC}$.

Problem 17.11: In this problem, we determine the number of points (a, b) such that $a^2 + b^2 = 25$ and the area of the triangle with vertices $(-5, 0)$, $(5, 0)$, and (a, b) is 10. (Source: MATHCOUNTS)

- What does the fact that (a, b) satisfies $a^2 + b^2 = 25$ tell us about where the point (a, b) is located on the coordinate plane?
- What does the fact that the area of the triangle with vertices $(-5, 0)$, $(5, 0)$ and (a, b) is 10 tell us about where the point (a, b) is located on the coordinate plane?
- How many points satisfy the conditions you found in parts (a) and (b)?

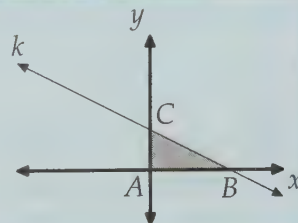
Problem 17.12: $ABCD$ is a rectangle with $AB = 8$ and $AD = 12$. Points X and Y are on $\overline{AB}$ and $\overline{CD}$, respectively, such that $AX = 7/3$ and $DY = 1$. $\overline{XY}$ meets diagonal $\overline{AC}$ at point P . In this problem, we find BP .

- Set the problem up on the Cartesian plane. Let A be the origin. What coordinates can we assign to each of B , C , and D ?
- What are the coordinates of P ?
- Find BP .

Problem 17.13: In $\triangle ABC$, we have $AB = BC = 8$ and $\angle ABC = 45^\circ$. In this problem, we find AC .

- Why should 45° make us think of the graph of $y = x$?
- What point on the line $y = x$ is 8 units from the origin?
- Let B be the origin. Find coordinates for A and C such that $\triangle ABC$ satisfies the problem. Find AC .

Problem 17.9: The area of the shaded region between line k and the axes in the diagram at right is 36. If line k passes through $(12, 0)$, then what is the slope of k ?



Solution for Problem 17.9: The shaded region is a right triangle. The vertex of the right angle of this triangle is the origin, $(0, 0)$, which is labeled A in the diagram. Vertex B is $(12, 0)$, so we know that leg $\overline{AB}$ has length 12. Because the area of the triangle is 36, we know that half the product of the lengths of the legs of the triangle is 36. Therefore, we have $(AB)(AC) = 72$, so $AC = 72/AB = 6$. So, vertex C is $(0, 6)$,

and the slope of k is $(6 - 0)/(0 - 12) = -1/2$. $\square$

Problem 17.10: Let A be $(-5, 6)$, B be $(-7, 9)$, and C be $(1, 10)$.

- (a) Find the length of the median of $\triangle ABC$ from B to $\overline{AC}$.
- (b) Find the length of the altitude of $\triangle ABC$ from A to $\overline{BC}$.

Solution for Problem 17.10:

- (a) The midpoint of $\overline{AC}$ is $(\frac{-5+1}{2}, \frac{6+10}{2}) = (-2, 8)$. The distance from B to this point is

$$\sqrt{[-2 - (-7)]^2 + (8 - 9)^2} = \sqrt{25 + 1} = \sqrt{26}.$$

- (b) *Solution 1: Find the coordinates of the foot of the altitude.* To find the length of the altitude from A to side $\overline{BC}$, we try to find the point P on $\overline{BC}$ such that $\overline{AP} \perp \overline{BC}$. First, we find the equation whose graph passes through B and C . The slope of this line is $(10 - 9)/[1 - (-7)] = 1/8$. The line passes through $(1, 10)$, so an equation of the line in point-slope form is $y - 10 = \frac{1}{8}(x - 1)$. Rearranging this equation gives $x - 8y = -79$.

Let point P be (x, y) . Since (x, y) is on $\overline{BC}$, it satisfies the equation $x - 8y = -79$. Because $\overline{AP} \perp \overline{BC}$, the product of the slopes of these two segments must be -1 . Since the slope of $\overline{BC}$ is $1/8$, we know the slope of $\overline{AP}$ is -8 . So, $\overline{AP}$ passes through (x, y) and $(-5, 6)$, and it has slope -8 . This gives us

$$\frac{y - 6}{x - (-5)} = -8.$$

Rearranging this equation gives $8x + y = -34$.

We now have the system of equations

$$\begin{aligned} x - 8y &= -79, \\ 8x + y &= -34. \end{aligned}$$

Multiplying the second equation by 8 then adding the result to the first equation gives us the equation $65x = -351$, which yields $x = -27/5$. Substituting this into either equation above then gives us $y = 46/5$. Therefore, $P = (-27/5, 46/5)$, so we have

$$AP = \sqrt{\left[-\frac{27}{5} - (-5)\right]^2 + \left(\frac{46}{5} - 6\right)^2} = \sqrt{\frac{4}{25} + \frac{256}{25}} = \sqrt{\frac{260}{25}} = \frac{2\sqrt{65}}{5}.$$

Solution 2: Use some geometric insight. The algebra in our first solution is a little messy. The system of equations we have to solve isn't very nice, and all the fractions give us plenty of opportunities to make mistakes. Once we see all that algebra coming our way, we might stop and think if there are any geometric insights we can use to simplify the problem.

Concept: When the algebra in an analytic geometry problem starts to get ugly, try using geometric facts to simplify the problem.

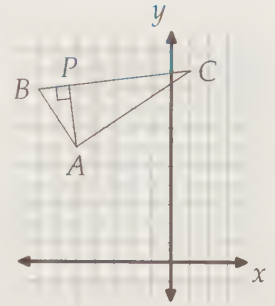


Here, we seek the length of the altitude from A to $\overline{BC}$. The altitude makes us think about area. We can easily find the length of $\overline{BC}$:

$$BC = \sqrt{(-7 - 1)^2 + (9 - 10)^2} = \sqrt{65}.$$

Therefore, if we can find $[ABC]$, we can easily find the length of $\overline{AP}$, since $(AP)(BC)/2 = [ABC]$.

Is $[ABC]$ easy to find? To answer this, we think about what kind of triangle ABC is. If it is isosceles or right, we can probably find the area easily. We can use slope to quickly check if it is right. We've already seen that the slope of $\overline{BC}$ is $1/8$. The slope of $\overline{AB}$ is $-3/2$ and the slope of $\overline{AC}$ is $2/3$. The product of the slopes of $\overline{AB}$ and $\overline{AC}$ is -1 , so $\overline{AB} \perp \overline{AC}$.



Concept: When given the coordinates of the vertices of a triangle, check if the triangle is special in any way. Most notably, check if the triangle is a right triangle. If it is, this fact will probably simplify the problem, because we know so much about right triangles.

Now the area of $\triangle ABC$ is easy to find. Because $\angle BAC = 90^\circ$, we have $[ABC] = (AB)(AC)/2$. Using the distance formula, we find that $AB = \sqrt{13}$ and $AC = 2\sqrt{13}$, so

$$[ABC] = \frac{(AB)(AC)}{2} = \frac{(\sqrt{13})(2\sqrt{13})}{2} = 13.$$

Finally, we have

$$AP = \frac{2[ABC]}{BC} = \frac{26}{\sqrt{65}} = \frac{26}{\sqrt{65}} \cdot \frac{\sqrt{65}}{\sqrt{65}} = \frac{2\sqrt{65}}{5}.$$

A little geometry sometimes goes a long way in analytic geometry problems!

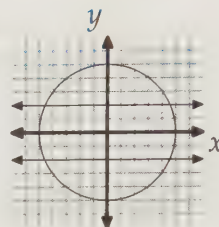
□

Problem 17.11: Find the number of points (a, b) such that $a^2 + b^2 = 25$ and the area of the triangle with vertices $(-5, 0)$, $(5, 0)$, and (a, b) is 10. (Source: MATHCOUNTS)

Solution for Problem 17.11: Let our triangle be PQR , with $P = (a, b)$, $Q = (-5, 0)$, and $R = (5, 0)$. Side $\overline{QR}$ is a horizontal segment with length 10. Because the area of $\triangle PQR$ is 10, we know that the length of the altitude from P to $\overline{QR}$ is 2. Since $\overline{QR}$ is along the x -axis, the fact that P is 2 units from $\overline{QR}$ means that P is either 2 units above or 2 units below the x -axis. Therefore, point P is on the line $y = -2$ or on the line $y = 2$. From here we present two solutions.

Solution 1: Algebra. Because P is on either $y = 2$ or $y = -2$, we must have either $b = 2$ or $b = -2$. When $b = 2$, the equation $a^2 + b^2 = 25$ gives us $a^2 = 25 - b^2 = 21$, so $a = \pm\sqrt{21}$. So, two possible points P are $(\sqrt{21}, 2)$ and $(-\sqrt{21}, 2)$. Similarly, letting $b = -2$ gives us two more points that satisfy the conditions in the problem, namely, $(\sqrt{21}, -2)$ and $(-\sqrt{21}, -2)$. Therefore, there are 4 different points that fit the description of the problem.

Solution 2: Geometry. Because (a, b) satisfies the equation $x^2 + y^2 = 25$, we know that (a, b) is on the circle centered at the origin with radius 5. We also know that (a, b) is either on the line $y = 2$ or the line $y = -2$. Each of these lines is 2 units from the center of the circle, so both lines go inside the circle. Therefore, both lines intersect the circle twice, as shown in the diagram at right. Each of the 4 points of intersection between a line and the circle gives us a point that satisfies the restrictions of the problem. $\square$



In our second solution, we interpreted the equation $a^2 + b^2 = 25$ as telling us that the point (a, b) is on the circle with center $(0, 0)$ and radius 5. Any time we see an equation that is in the standard form of a circle, we might consider using analytic geometry. This gives us a new tool for tackling algebraic problems:

Concept: Interpreting an equation as a geometric figure on the Cartesian plane can often help us solve problems.



This street goes both ways! Sometimes it is useful to put a geometry problem on the Cartesian plane and use algebraic tools to solve the problem.

Problem 17.12: $ABCD$ is a rectangle with $AB = 8$ and $AD = 12$. Points X and Y are on $\overline{AB}$ and $\overline{CD}$, respectively, such that $AX = 7/3$ and $DY = 1$. $\overline{XY}$ meets diagonal $\overline{AC}$ at point P . Find BP .

Solution for Problem 17.12: We seek a length, so we might try building right triangles and using the Pythagorean Theorem. (Try doing so on your own.) Analytic geometry offers us another way to find lengths: if we can find the coordinates of two points, we can use the distance formula to find the distance between them. So, we try turning this problem into an analytic geometry problem. One reason we do so is that rectangles are particularly easy to describe with coordinates.

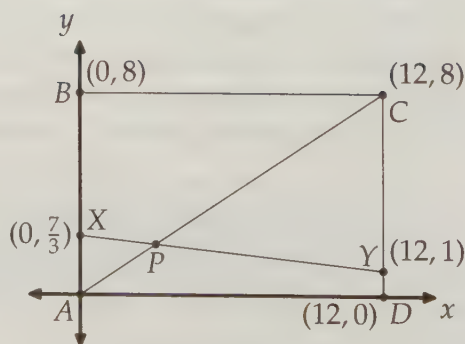
We start by letting A be the origin, and letting two sides of the rectangle be along the coordinate axes. This makes a lot of the coordinates in the problem equal to 0, and 0's are easy to deal with.

Concept: When setting a geometry problem up on the Cartesian plane, choose the origin and the axes in a convenient matter. Often this means letting the origin be the vertex of a right angle in the problem, so that the sides of the angle are along the axes.



In the diagram at right, we let $\overline{AB}$ be along the y -axis and $\overline{AD}$ be along the x -axis. Because $AB = 8$ and $AD = 12$, point B is $(0, 8)$ and point D is $(12, 0)$. Because X is on $\overline{AB}$ such that $AX = 7/3$, we have $X = (0, 7/3)$. Similarly, Y is 1 unit above D , so $Y = (12, 1)$.

Now, if we can find the coordinates of point P , we can use the distance formula to find BP . We know that P is on $\overline{AC}$. Because $\overrightarrow{AC}$ has slope $(8 - 0)/(12 - 0) = 2/3$ and it passes through the origin, the equation whose graph is $\overline{AC}$ is $y = 2x/3$. (This is why we like having so many 0's in our coordinates – the resulting equations



for lines connecting points in the diagram are simpler.) Point P is also on $\overline{XY}$. The slope of $\overrightarrow{XY}$ is $(1 - 7/3)/(12 - 0) = -1/9$ and its y -intercept is $(0, 7/3)$, so the slope-intercept form of the equation of this line is

$$y = -\frac{1}{9}x + \frac{7}{3}.$$

Rearranging this equation gives $x + 9y = 21$. So, P is the intersection of the graphs of $y = 2x/3$ and $x + 9y = 21$. Substituting $y = 2x/3$ into $x + 9y = 21$ gives $x + 6x = 21$, from which we find $x = 3$. Therefore, we have $y = 2x/3 = 2$, so P is $(3, 2)$. Finally, because B is $(0, 8)$, we have

$$BP = \sqrt{(3 - 0)^2 + (2 - 8)^2} = \sqrt{9 + 36} = \sqrt{45} = 3\sqrt{5}.$$

A little algebra sometimes goes a long way in geometry problems! $\square$

WARNING!!



Don't get carried away with using analytic geometry to solve geometry problems. While our solution to Problem 17.12 was pretty straightforward with analytic geometry, you'll find that most geometry problems are easier to solve with geometric methods than with analytic geometry.

For example, suppose we wished to find the length of the median to the hypotenuse of a right triangle with sides of length 14, 48, and 50. It is much faster to use the simple geometric fact that the median to the hypotenuse of a right triangle is half the length of the hypotenuse than it is to assign coordinates to the vertices of the triangle, find the coordinates of the midpoint of the hypotenuse, then use the distance formula.

After finishing this chapter, try flipping back through this book and finding problems that you can solve with analytic geometry. You'll find some common trends among the ones you can solve, such as rectangles, midpoints, and right triangles. You'll also find some trends among problems that are hard to solve with analytic geometry, such as circles, angles that are not 45° or 90° , and complicated diagrams.

Our last sentence suggests that problems with 45° angles might be candidates for analytic geometry. Let's see why.

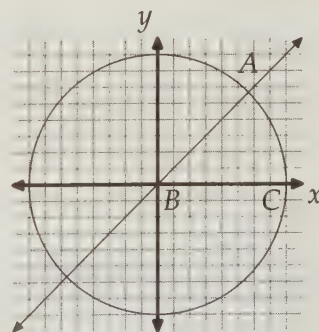
Problem 17.13: In $\triangle ABC$, we have $AB = BC = 8$ and $\angle ABC = 45^\circ$. In this problem, we find AC .

Solution for Problem 17.13: We present both an analytic geometry solution and a 'pure geometry' solution.

Solution 1: Analytic geometry. The 45° angle might make us think about analytic geometry because the graph of the line $y = x$ makes a 45° angle with the coordinate axes. We also notice that $AB = BC = 8$ means that A and C are both on the circle with center B and radius 8. In other words, if we let B be the origin, then the coordinates of A and C are solutions to the equation

$$x^2 + y^2 = 64,$$

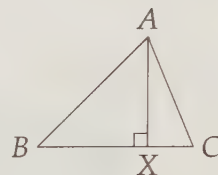
because the graph of this equation is the circle with center $(0, 0)$ and radius 8. We know that the graph of $y = x$ makes a 45° angle with the axes, so we let C be along the x -axis, at $(8, 0)$, and A be on the graph of the line $y = x$.



We can find the coordinates of A either by drawing an altitude from A to the x -axis, or by using our equation for the circle. Because A is on both the circle and the graph of $y = x$, we can find the coordinates of A by substituting $y = x$ into our equation for the circle. This gives us $x^2 + x^2 = 64$, from which we find $x = 4\sqrt{2}$. (In our diagram, A is on the right side of the y -axis, so x must be positive.) Therefore, point A is $(4\sqrt{2}, 4\sqrt{2})$, and we can use the distance formula to find

$$AC = \sqrt{(4\sqrt{2} - 8)^2 + (4\sqrt{2} - 0)^2} = \sqrt{32 - 64\sqrt{2} + 64 + 32} = \sqrt{128 - 64\sqrt{2}} = 8\sqrt{2 - \sqrt{2}}.$$

Solution 2: Geometry. Our analytic geometry solution might inspire a purely geometric solution. At the point where we were finding the coordinates for A , we suggested drawing an altitude from A to $\overline{BC}$. Suppose we do that right from the beginning. (We might also be inspired to do this by the 45° angle; drawing this altitude builds a 45-45-90 triangle).



From 45-45-90 triangle ABX , we have $AX = XB = AB/\sqrt{2} = 4\sqrt{2}$. Therefore, we have $CX = CB - BX = 8 - 4\sqrt{2}$. We can then apply the Pythagorean Theorem to $\triangle AXC$ to find $AC = 8\sqrt{2 - \sqrt{2}}$, as before. $\square$

Concept: Solving a problem in two ways is a good way to check your answer.



Our two solutions in the last problem also show that analytic geometry solutions and purely geometric solutions to the same problem are often closely related to each other.

Exercises

17.3.1 In Problem 17.10, we used slopes to determine that $\triangle ABC$ is a right triangle. How else could we have determined that $\triangle ABC$ is a right triangle?

17.3.2 Find the area of the triangle bounded by the graphs of the equations $y = x$, $y = -x$, and $y = 6$. (Source: AMC 12)

17.3.3 What is the shortest possible distance between a point on the graph of $x^2 + y^2 + 6x - 8y = 0$ and a point on the graph of $x^2 - 14x + y^2 + 10y + 65 = 0$?

17.3.4 In $\triangle ABC$, we have $AB = 8$, $BC = 12$, and $\angle ABC = 30^\circ$. Find AC .

17.3.5★ Find the radius of each circle that passes through $(9, 2)$ and is tangent to both the x -axis and the y -axis. **Hints:** 7, 487

17.4 Proofs with Analytic Geometry

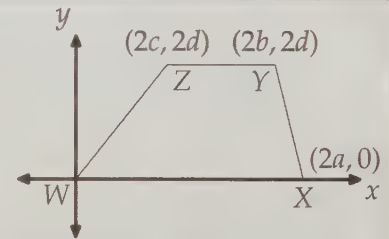
In this section, we learn how to prove geometric facts using analytic geometry.

Problems

Problem 17.14: In this problem, we prove that the midpoint of the hypotenuse of a right triangle is the circumcenter of the triangle.

- Let the right triangle be $\triangle ABC$, with right angle at C . Why is it useful to choose C to be the origin of the Cartesian plane?
- Suppose C is the origin. Why should we choose the axes of the Cartesian plane such that the legs of $\triangle ABC$ are along the axes?
- Let A be $(a, 0)$ and B be $(0, b)$. What is the midpoint of the hypotenuse?
- Show that the midpoint of the hypotenuse is equidistant from the vertices of the triangle.

Problem 17.15: In this problem, we prove that the median of a trapezoid is parallel to the bases of the trapezoid, and that the length of the median is the average of the lengths of the bases. Let the trapezoid be $WXYZ$, where W is the point $(0, 0)$, X is the point $(2a, 0)$, Y is the point $(2b, 2d)$, Z is the point $(2c, 2d)$, and we have $a > 0$, $b > c$, and $d > 0$.



- Why is $WXYZ$ a trapezoid? Which two sides are parallel?
- Find the coordinates of the midpoints of the legs of $WXYZ$. Why did we use $2a$, $2b$, $2c$, and $2d$ to describe the coordinates of the vertices of $WXYZ$, instead of using just a , b , c , and d ?
- Show that the median of $WXYZ$ is parallel to the bases of $WXYZ$, and equal in length to the average of the lengths of the bases.
- Explain why our proof is valid by describing why for every trapezoid there is some choice of origin and axes such that the vertices of the trapezoid can be described with the coordinates $(0, 0)$, $(2a, 0)$, $(2b, 2d)$, and $(2c, 2d)$.

Problem 17.16: In this problem we prove that the medians of a triangle are concurrent. Let our triangle be ABC , where $A = (0, 0)$, $B = (2a, 0)$, and $C = (2b, 2c)$.

- Find the coordinates of the midpoint of $\overline{AB}$. Find an equation whose graph is the line containing the median from C to $\overline{AB}$.
- Find an equation of the line whose graph is the line containing the median from B to $\overline{AC}$.
- Find the intersection of the lines you found in parts (a) and (b).
- Prove that the intersection point you found in part (c) is on the median from A to $\overline{BC}$.

Problem 17.17: Suppose line k divides rectangle $ABCD$ into two pieces of equal area. Prove that line k passes through the intersection of the diagonals of $ABCD$.

Problem 17.14: Use analytic geometry to prove that the midpoint of the hypotenuse of any right triangle is the circumcenter of the triangle.

Solution for Problem 17.14: Our first step is to set up the problem on the Cartesian plane. We must be very careful when we do so. See if you can figure out why the set-up in the Bogus Solution below is not sufficient:

Bogus Solution:

We let our triangle be ABC , and we let $C = (0, 0)$, $A = (4, 0)$, and $B = (0, 6)$. Legs $\overline{AC}$ and $\overline{BC}$ are along the axes, so they are perpendicular. Therefore, $\triangle ABC$ is a right triangle with hypotenuse $\overline{AB}$. The midpoint of $\overline{AB}$ is $(2, 3)$. Let this midpoint be M . We have

$$AM = \sqrt{(2 - 4)^2 + (3 - 0)^2} = \sqrt{13},$$

$$BM = \sqrt{(2 - 0)^2 + (3 - 6)^2} = \sqrt{13},$$

$$CM = \sqrt{(2 - 0)^2 + (3 - 0)^2} = \sqrt{13}.$$

Since M is equidistant from the vertices of $\triangle ABC$, point M is the circumcenter of $\triangle ABC$.

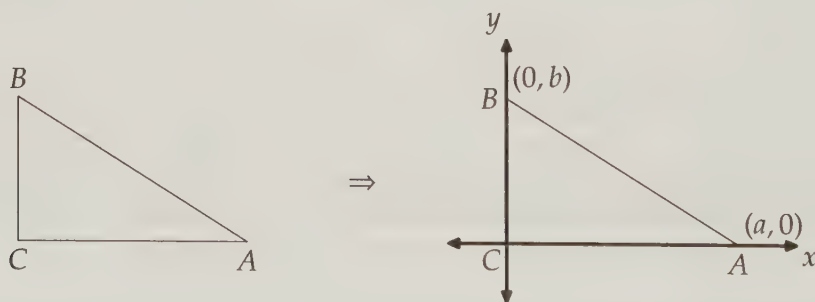
This does indeed show why the midpoint of the hypotenuse of this particular right triangle is the circumcenter of this right triangle. However, what about other right triangles with different side lengths? Our 'proof' doesn't cover every possible right triangle.

WARNING!!

When setting up a proof on the Cartesian plane, we must be very careful. Our proof must address all possible configurations of the problem, so our analytic geometry representation of the problem must cover all possible configurations.

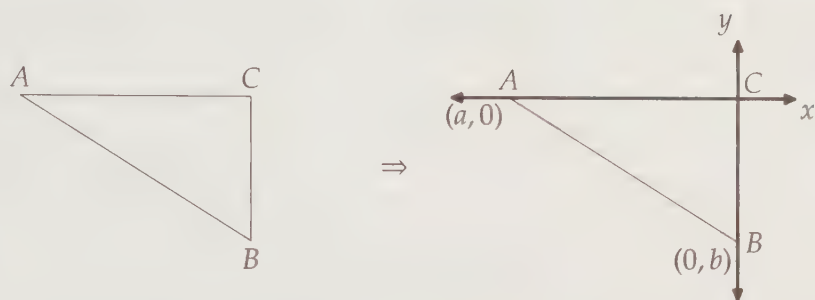
We might start by letting $A = (a, b)$, $B = (c, d)$, and $C = (e, f)$. But that's six variables! Hopefully we can find a simpler representation of $\triangle ABC$.

When setting up a geometry problem on the Cartesian plane, we can start with a diagram, then add the coordinate axes. We try to do so in a way that simplifies our problem. We can choose any point to be the origin, so we choose one of the vertices of the right triangle to be our origin.



We choose the vertex of the right angle, point C , to be the origin because this allows us to choose our axes so that the legs of the right triangle are along the axes. So, we let $A = (a, 0)$ and $B = (0, b)$.

Notice that a and/or b could be positive or negative. For example, in the case below, both a and b are negative.



Since C is the vertex of the right angle, side $\overline{AB}$ is the hypotenuse. Let M be the midpoint of $\overline{AB}$, so that

$$M = \left(\frac{a}{2}, \frac{b}{2} \right).$$

To show that M is the circumcenter of $\triangle ABC$, we must show that M is equidistant from all three vertices. This is a job for the distance formula:

$$AM = \sqrt{\left(\frac{a}{2} - a\right)^2 + \left(\frac{b}{2} - 0\right)^2} = \sqrt{\frac{a^2 + b^2}{4}},$$

$$BM = \sqrt{\left(\frac{a}{2} - 0\right)^2 + \left(\frac{b}{2} - b\right)^2} = \sqrt{\frac{a^2 + b^2}{4}},$$

$$CM = \sqrt{\left(\frac{a}{2} - 0\right)^2 + \left(\frac{b}{2} - 0\right)^2} = \sqrt{\frac{a^2 + b^2}{4}}.$$

We have $AM = BM = CM$, so M is the circumcenter of $\triangle ABC$.

Notice that every step in our proof is valid even if a or b or both are negative. $\square$

Important:



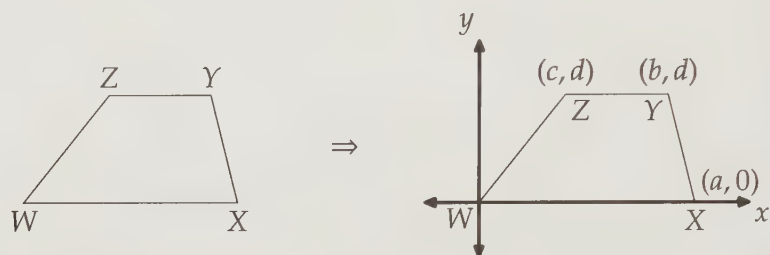
When setting up a geometry problem on the Cartesian plane, choose your origin and your coordinate axes wisely. Typically, we do so in a way that makes the coordinates of important points in the problem as simple as possible.

Notice that our proof in Problem 17.14 only applies to right triangles, not to all triangles. The points we chose to represent the vertices, $(0, 0)$, $(a, 0)$, and $(0, b)$, are always the vertices of a right triangle (when a and b are nonzero). We cannot use these three points as our vertices to prove a fact about all triangles, since any triangle with vertices $(0, 0)$, $(a, 0)$, and $(0, b)$ must be a right triangle.

Problem 17.15: Use analytic geometry to prove that the median of a trapezoid is parallel to the bases of the trapezoid, and that the length of the median is the average of the lengths of the bases.

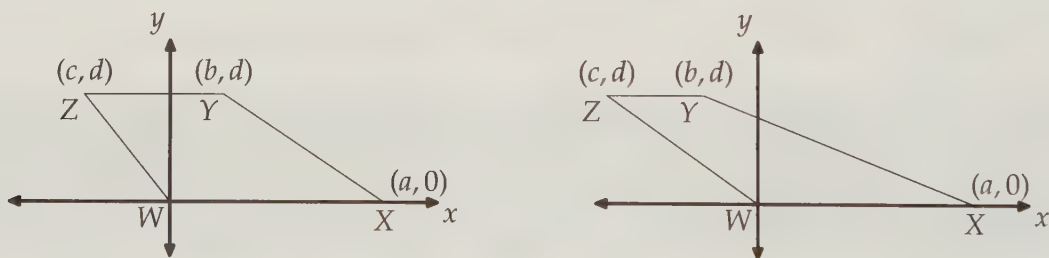
Solution for Problem 17.15: On page 211, we saw geometric proofs of these facts. Here, we try to find an analytic geometry proof. We might start by letting the vertices of our trapezoid be (a, b) , (c, d) , (e, f) , and (g, h) . Yikes. Eight variables! Maybe we can do better.

As noted in Problem 17.14, we can start with a diagram, then add the coordinate axes to set up a proof on the Cartesian plane. Here, we choose the x -axis to contain one of the bases such that the other base is above the x -axis. We then choose the leftmost of the two vertices on the x -axis to be the origin. So, one vertex is the origin and the other is $(a, 0)$ with $a > 0$. We'll call the origin W and the second vertex on the x -axis X , as shown below.



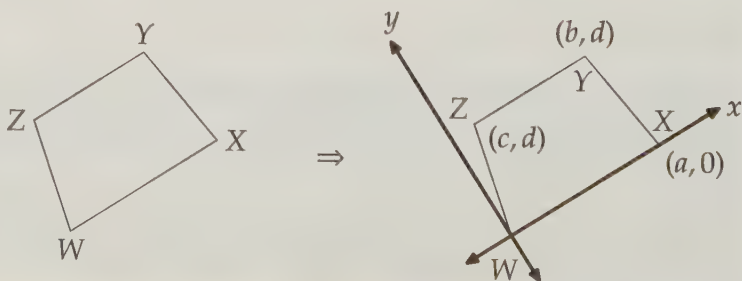
We placed the x -axis such that the endpoints of one base are on the x -axis because this forces the endpoints of the other base to have the same y -coordinate, since the bases are parallel. By choosing the 'lower' base to be x -axis, we force this common y -coordinate to be positive. Therefore, the other two vertices of the trapezoid can be represented by (b, d) and (c, d) , where $d > 0$ and $b > c$. Finally, we let Y be (b, d) and Z be (c, d) , so $\overline{XY}$ and $\overline{WZ}$ are the legs of the trapezoid. Now, we have represented our trapezoid with only 4 variables, instead of 8.

Notice that every trapezoid can be represented in this way. Because b and c can take on negative values, letting the vertices be $W = (0, 0)$, $X = (a, 0)$, $Y = (b, d)$, and $Z = (c, d)$ with $a > 0$, $d > 0$, and $b > c$ will also represent trapezoids in which Z and/or Y end up to the left of the y -axis. Two examples are shown below. In each case, we have W at the origin, X to the right of W on the x -axis, and $\overline{YX}$ and $\overline{ZW}$ as the legs (because $b > c$).

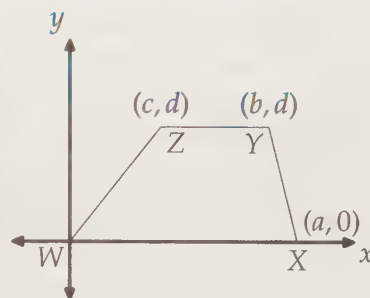


But what about trapezoids in which the bases are not horizontal?

There's no reason we have to make our x -axis and y -axis horizontal and vertical! They only have to be perpendicular. For example, consider the trapezoid with 'slanted' bases at right. We similarly make our axes 'slanted' so that the x -axis includes one base.



Now, we're ready for our proof. We'll use the diagram in which $\overline{YZ}$ is horizontal and both Y and Z are to the right of the y -axis, as shown at right. (However, our proof will be valid for all set-ups we have discussed.) The midpoint of leg $\overline{WZ}$ is $(\frac{c}{2}, \frac{d}{2})$ and the midpoint of leg $\overline{XY}$ is $(\frac{a+b}{2}, \frac{d}{2})$. The median connects these two points, so the median is on the line $y = \frac{d}{2}$. Therefore, the median is horizontal, so it is parallel to the bases. We can also use the coordinates of the midpoints of the legs to determine that the length of the median is



$$\frac{a+b}{2} - \frac{c}{2} = \frac{a+b-c}{2}.$$

The lengths of the bases are $WX = a$ and $YZ = b - c$, so the average of the lengths of the bases is also $(a + b - c)/2$. So, the length of the median equals the average of the lengths of the bases. Notice that every step of this proof is valid for all the other arrangements we showed.

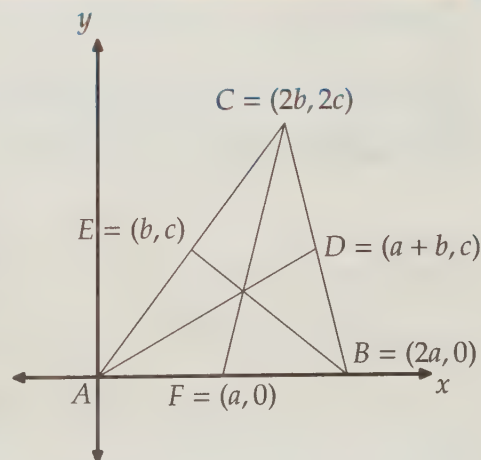
We also could have avoided the fractions in our solution by being a little clever about assigning variables to coordinates. Because we are going to have to work with midpoints to get information about the median, we know we'll have to divide expressions by 2. Therefore, we might make our coordinates $W = (0, 0)$, $X = (2a, 0)$, $Y = (2b, 2d)$, and $Z = (2c, 2d)$. Then, the endpoints of the median are $(a + b, d)$ and (c, d) . No fractions! See if you can finish the problem from here. $\square$

Concept: When we set up a geometry problem involving midpoints on the Cartesian plane, we often use $2a$, $2b$, $2c$, etc. for coordinates, rather than just a , b , c , etc. This helps us avoid expressions involving fractions when we find midpoints of segments in the problem.

Speaking of midpoints, these are often a sign that an analytic geometry approach might work. This is because coordinates of midpoints are particularly easy to find.

Problem 17.16: Prove that the medians of any triangle are concurrent.

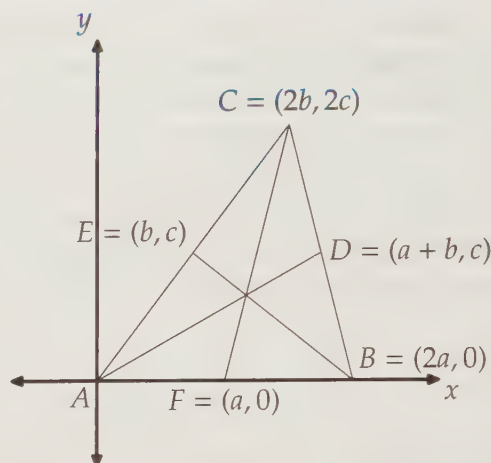
Solution for Problem 17.16: We must prove a statement about medians, so we're working with midpoints. This might make us consider analytic geometry. First, we have to set up the problem on the Cartesian plane. Let $\triangle ABC$ be our triangle, and let the medians be $\overline{AD}$, $\overline{BE}$, and $\overline{CF}$. We let A be the origin and choose our axes so that B is on the x -axis, as shown at right. Since we will have to find the coordinates of the midpoints of the sides of the triangle, we let A be $(0, 0)$ and B be $(2a, 0)$. We have no restrictions on C except that it cannot be on the x -axis, so we let C be $(2b, 2c)$, where $c \neq 0$. Again, we use $2b$ and $2c$ because we will be finding the midpoints of the sides of the triangle. Specifically, the midpoints of the sides of $\triangle ABC$ are $D = (a + b, c)$, $E = (b, c)$, and $F = (a, 0)$.



Now that we have the coordinates of two points on each median, we can find the linear equations whose graphs include the medians. We then must show that there is a point (x, y) that satisfies all three equations. First, we'll find the equation for $\overleftrightarrow{AD}$. The slope of $\overleftrightarrow{AD}$ is $c/(a+b)$ and $\overleftrightarrow{AD}$ passes through $(0, 0)$, so an equation whose graph is $\overleftrightarrow{AD}$ is

$$y - 0 = \frac{c}{a+b}(x - 0).$$

Notice that the right side of this equation is undefined if $a+b=0$. Since the slope of $\overleftrightarrow{AD}$ is $c/(a+b)$, we only have $a+b=0$ if $\overleftrightarrow{AD}$ is a vertical line. Specifically, if $a+b=0$, then A and D are both on the y -axis.



WARNING!! When using analytic geometry for a proof, be extra careful not to overlook special cases.

To take care of this special case, we multiply both sides of the equation by $a+b$ to get

$$(a+b)y = c(x-0).$$

If $\overleftrightarrow{AD}$ is vertical, then $a+b=0$, and $\overleftrightarrow{AD}$ is the graph of $x=0$. Our equation above becomes $x=0$ when $a+b=0$, so this equation describes $\overleftrightarrow{AD}$ even if $\overleftrightarrow{AD}$ is a vertical line.

Notice that we use $(0,0)$ when writing a point-slope form of the equation of $\overleftrightarrow{AD}$ instead of using $(a+b, c)$. This makes our equation considerably simpler, and is one of the reasons we like to make as many of our coordinates 0 as possible.

Concept: Don't make analytic geometry proofs harder than they need to be!

Similarly, the slope of $\overleftrightarrow{BE}$ is $c/(b-2a)$ and $\overleftrightarrow{BE}$ passes through $(2a, 0)$, so an equation whose graph is $\overleftrightarrow{BE}$ is

$$y - 0 = \frac{c}{b-2a}(x - 2a).$$

Multiplying both sides by $b-2a$ gives

$$(b-2a)y = c(x-2a).$$

If $b=2a$, then $\overleftrightarrow{BE}$ is the graph of $x=2a$, since $\overleftrightarrow{BE}$ must pass through $(2a, 0)$ and (b, c) . If $b=2a$, then $(b-2a)y = c(x-2a)$ becomes $0 = c(x-2a)$, which simplifies to $x=2a$. So, the equation $(b-2a)y = c(x-2a)$ describes $\overleftrightarrow{BE}$ even if $\overleftrightarrow{BE}$ is a vertical line.

And finally, the slope of $\overleftrightarrow{CF}$ is $2c/(2b-a)$ and $\overleftrightarrow{CF}$ passes through $(a, 0)$, so for this line we have

$$y - 0 = \frac{2c}{2b-a}(x - a).$$

Again, we multiply both sides by the denominator of the slope to get

$$(2b - a)y = 2c(x - a),$$

which describes $\overleftrightarrow{CF}$ even if $\overleftrightarrow{CF}$ is a vertical line.

We now have the three equations

$$\begin{aligned}\overleftrightarrow{AD}: \quad y(a + b) &= xc, \\ \overleftrightarrow{BE}: \quad y(b - 2a) &= xc - 2ac, \\ \overleftrightarrow{CF}: \quad y(2b - a) &= 2xc - 2ac.\end{aligned}$$

We can find the coordinates of the intersection of the graphs of the first two equations by substituting $xc = y(a + b)$ into $y(b - 2a) = xc - 2ac$. This gives us

$$y(b - 2a) = y(a + b) - 2ac.$$

Subtracting $y(a + b)$ from both sides gives us $y(b - 2a) - y(a + b) = -2ac$. Simplifying the left side then gives $-3ay = -2ac$. We know that $a \neq 0$, so we can divide by $-3a$ to find $y = 2c/3$. Since we also have $xc = y(a + b)$, we find $x = y(a + b)/c = 2(a + b)/3$.

So, we know that $\overline{AD}$ and $\overline{BE}$ meet at $(\frac{2(a+b)}{3}, \frac{2c}{3})$. All we have left is to see if this point is on $\overline{CF}$. We do so by substituting $x = 2(a + b)/3$ and $y = 2c/3$ into our equation for $\overleftrightarrow{CF}$ from above. Substituting these expressions into $y(2b - a) = 2xc - 2ac$ gives us a left side of

$$y(2b - a) = \frac{2c}{3}(2b - a) = \frac{4bc}{3} - \frac{2ac}{3},$$

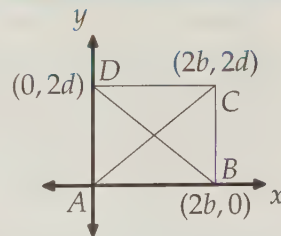
and a right side of

$$2xc - 2ac = 2 \left[\frac{2(a + b)}{3} \right] c - 2ac = \frac{4ac}{3} + \frac{4bc}{3} - 2ac = \frac{4bc}{3} - \frac{2ac}{3}.$$

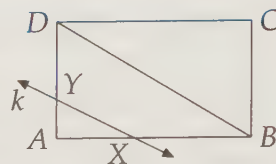
We see that the two sides are the same for the ordered pair $(x, y) = (\frac{2(a+b)}{3}, \frac{2c}{3})$, so this point is on our third median, as well! Therefore, we can conclude that the medians of a triangle are concurrent. $\square$

Problem 17.17: Suppose line k divides rectangle $ABCD$ into two pieces of equal area. Prove that line k passes through the intersection of the diagonals of $ABCD$.

Solution for Problem 17.17: It's quite easy to describe a general rectangle with coordinates. We let A be the origin and choose the axes so that two sides are along the axes. Because $ABCD$ is a rectangle, it is also a parallelogram, so its diagonals bisect each other. This means we want to prove that k passes through the midpoint of $\overline{AC}$. Therefore, we let $B = (2b, 0)$ and $D = (0, 2d)$ (instead of using $B = (b, 0)$ and $D = (0, d)$). Point C is on the vertical line through B and on the horizontal line through D , so $C = (2b, 2d)$. This means the midpoint of $\overline{AC}$ is (b, d) .



Having dealt with $ABCD$, it's time to take care of k . First, we have to think about ways in which k could intersect the rectangle. If k intersects two consecutive sides (excluding the vertices of the rectangle), then the area of the triangle it forms is smaller than the area of the triangle formed by drawing a diagonal. For example, in the diagram at right, we have $[AXY] < [ABD] = [ABCD]/2$. So, it is impossible for such a line to bisect the area of $ABCD$.



Following similar logic, if k passes through a vertex of $ABCD$ and bisects the area of $ABCD$, then line k must contain a diagonal of $ABCD$. So, line k clearly passes through the intersection point of the diagonals in this case.

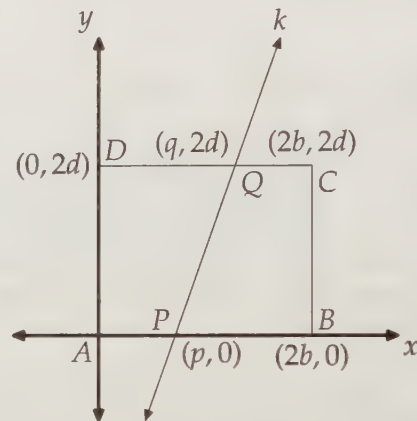
The other possibility is that k intersects opposite sides of the rectangle (excluding the vertices). Shown at right is our set-up on the Cartesian plane. Line k intersects $\overline{AB}$ and $\overline{CD}$ at $P = (p, 0)$ and $Q = (q, 2d)$ as shown. If k bisects the area of $ABCD$, then it splits it into two trapezoids with equal area. The bases of trapezoid $APQD$ have lengths p and q , and $APQD$ has height $2d$, so we have

$$[APQD] = (2d) \left(\frac{p+q}{2} \right) = d(p+q).$$

The bases of $PBCQ$ have lengths $2b - p$ and $2b - q$, so we have

$$[PBCQ] = (2d) \left(\frac{2b-p+2b-q}{2} \right) = d(4b-p-q).$$

Setting these equal gives us $d(p+q) = d(4b-p-q)$. Dividing by d gives us $p+q = 4b-p-q$, so $2(p+q) = 4b$. Finally, we find that $p+q = 2b$. (We also could have used the fact the $[APQD] = \frac{1}{2}[ABCD]$ to show that $p+q = 2b$.)



But how does this help? We want to show that $\overleftrightarrow{PQ}$ passes through the midpoint of $\overline{AC}$. The midpoint of $\overline{AC}$ is (b, d) . Now that we know that $p+q = 2b$, we see that the midpoint of $\overline{PQ}$ is

$$\left(\frac{p+q}{2}, \frac{0+2d}{2} \right) = (b, d).$$

Therefore, line k passes through the intersection of the diagonals of $ABCD$. $\square$

Exercises

17.4.1 In part (a) of Problem 17.4.2, we will use analytic geometry to prove a fact about parallelograms. Which of the following groups of vertices can we use to represent the vertices of the parallelogram, to prove a fact for all parallelograms? If a group cannot be used, describe a parallelogram whose vertices cannot be represented by the given points. You can assume all variables represent nonzero numbers.

For example, if we wished to prove a result about all triangles, we cannot use $(0, 0)$, $(a, 0)$, and $(0, b)$ to represent the vertices of the triangle. This is because any triangle with these points as vertices is a right triangle. So, these points cannot represent the vertices of any triangle that is not a right triangle.

(a) $(0, 0), (a, 0), (b, b), (b, c)$

(c) $(0, 0), (b, b), (b+a, b), (a, 0)$

(b) $(a, 0), (0, b), (-a, 0), (0, -b)$

(d) $(0, 0), (b-a, c), (b, c), (a, 0)$

17.4.2

- (a) Use analytic geometry to prove that the diagonals of a parallelogram bisect each other.
- (b)★ Use analytic geometry to prove that if the diagonals of a quadrilateral bisect each other, then the quadrilateral is a parallelogram.

17.4.3 Use analytic geometry to show that if the diagonals of a quadrilateral are perpendicular and bisect each other, then the quadrilateral is a rhombus.

17.4.4★ Notice that in Problem 17.16, the sum of the equations we found for $\overleftrightarrow{AD}$ and $\overleftrightarrow{BE}$ is the equation for $\overleftrightarrow{CF}$. Why could we have used this observation to deduce that $\overleftrightarrow{AD}$, $\overleftrightarrow{BE}$, and $\overleftrightarrow{CF}$ are concurrent?

17.5 Distance Between a Point and a Line

The distance between a point and a line is the shortest distance from the point to a point on the line.

Problems

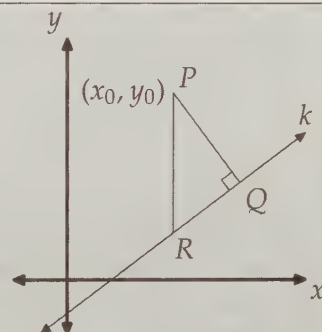
Problem 17.18: Find the distance between the point $(2, 7)$ and the graph of the line $x - 3y = 1$.

Problem 17.19: In this problem, we find a formula for the distance between the point (x_0, y_0) and the line $Ax + By + C = 0$, where A, B, C, x_0 , and y_0 are all constants. We'll find a formula assuming that $AB \neq 0$; you'll be asked to show that this formula also works when $AB = 0$ as an Exercise.

Let k be the line, P be the point (x_0, y_0) , and Q be the foot of the altitude from P to k . Let R be the point on k that has x -coordinate x_0 .

For the following parts, assume P is above line k on the Cartesian plane, and that k has a positive slope. (The proofs for all other possibilities are essentially the same.)

- (a) Explain why $QR/PQ = -A/B$.
- (b) Explain why $PR = (Ax_0 + By_0 + C)/B$. (Hint: Remember that R is on k , and has the same x -coordinate as P . By how much do we have to increase the y -coordinate of R to get the y -coordinate of P ?)
- (c) Show that the distance from point P to line k is $\frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}$.



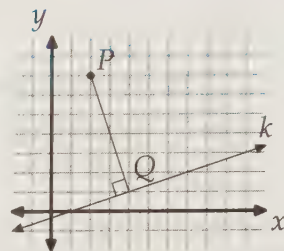
Problem 17.20: Use your formula from Problem 17.19 to check your answer to Problem 17.18.

Problem 17.21: How many ordered pairs (x, y) satisfy both $14x - 48y + 49 = 0$ and $x^2 + y^2 = 1$?

We start by finding the distance between a specific point and a specific line.

Problem 17.18: Find the distance between the point $(2, 7)$ and the graph of the line $x - 3y = 1$.

Solution for Problem 17.18: Let point P be $(2, 7)$ and let line k be the graph of the line $x - 3y = 1$. Let Q be the point on k that is closest to P , so we have $\overrightarrow{PQ} \perp k$. Therefore, the product of the slopes of these lines is -1 . Writing $x - 3y = 1$ in slope-intercept form gives $y = x/3 - 1/3$, so the slope of k is $1/3$. Therefore, the slope of $\overrightarrow{PQ}$ is -3 . Since Q is on the line through $(2, 7)$ with slope -3 , it is on the graph of $y - 7 = -3(x - 2)$. Rearranging this equation gives $3x + y = 13$.



We now have two equations whose graphs pass through point Q , so we find the coordinates of Q by solving the system of equations

$$\begin{aligned}x - 3y &= 1, \\3x + y &= 13.\end{aligned}$$

Solving this system of equations gives $(x, y) = (4, 1)$, so point Q is $(4, 1)$. As a quick check, we note that the slope of $\overrightarrow{PQ}$ is $(1 - 7)/(4 - 2) = -3$, so $\overrightarrow{PQ}$ is indeed perpendicular to k . (If this slope had not come out to -3 , then we would have known we made a mistake.)

Concept: Checking intermediate steps while working on problems will help you catch a lot of errors.



Now, we use the distance formula to find

$$PQ = \sqrt{(2 - 4)^2 + (7 - 1)^2} = \sqrt{4 + 36} = 2\sqrt{10}.$$

□

We could follow the procedure we used in Problem 17.18 to find the distance between any point and any line. However, the procedure requires a lot of steps, and often the numbers will get pretty ugly. Instead, let's see if we can find a formula to use to find the distance between a point (x_0, y_0) and the line that is the graph $Ax + By + C = 0$.

Problem 17.19: Find a formula for the distance between the point (x_0, y_0) and the line $Ax + By + C = 0$, where A, B, C, x_0 , and y_0 are all constants.

Solution for Problem 17.19: Let P be the point (x_0, y_0) and k be the graph of $Ax + By + C = 0$. We could proceed as we did in our solution to Problem 17.18, by finding the slope of k , then finding the equation of the line through P that is perpendicular to k . Then, we find the intersection of k and this new line, and then...

That looks like a lot of work. Before diving into pages of algebra, let's see if we can use some geometric insights to simplify the problem.

Concept: Algebra is not the only tool we have to solve problems about the Cartesian plane. Combining geometric insights with algebra can lead to very nice solutions.



We start by drawing a diagram with P , k , and the perpendicular segment from P to k . Let point Q be the foot of the perpendicular from P to k . We seek a length, and we have a right angle, so we think about building right triangles. We build a right triangle by drawing a vertical segment from P to line k , meeting k at point R , as shown.

Since P and R are on the same vertical line, the x -coordinate of R is x_0 . Since R is on k , its coordinates satisfy the equation $Ax + By + C = 0$. We can now use this equation to find the y -coordinate of R . Let y_R be the y -coordinate of R , so R is (x_0, y_R) . Because R is on k , we must have

$$Ax_0 + By_R + C = 0.$$

Solving for y_R gives us

$$y_R = \frac{-Ax_0 - C}{B}.$$

It's still not clear how we can find the coordinates of Q . Right triangles make us think of similar triangles. Drawing the altitude $\overline{QS}$ to the hypotenuse of right triangle $\triangle PQR$ gives us plenty of similar triangles.

But what good are they? We haven't used any information about the equation of our line yet, so we focus on that. When we do so, we see that we can relate lengths of segments in our diagram to the slope of k . Specifically, we see that the slope of line k equals SR/QS . (In our diagram, we assume the slope of k is positive. Essentially the same approach will work if the slope of k is negative.) We can also write the slope of k in terms of A and B . Putting the equation $Ax + By + C = 0$ in slope-intercept form gives

$$y = -\frac{A}{B}x - \frac{C}{B},$$

so the slope of k is $-A/B$. Therefore, we have

$$\frac{SR}{QS} = -\frac{A}{B}.$$

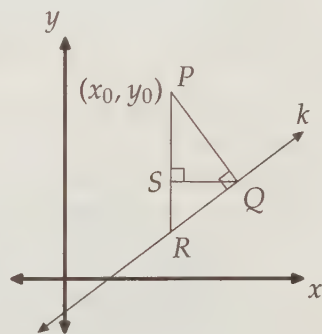
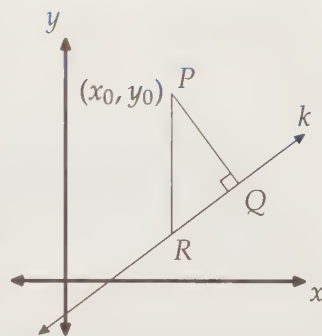
Our similar triangles give us a way to relate $\triangle QSR$ to $\triangle PQR$. We have $\triangle QSR \sim \triangle PQR$, so

$$\frac{SR}{QS} = \frac{QR}{PQ}.$$

Combining this with $SR/QS = -A/B$ gives us $QR/PQ = -A/B$. Rearranging this gives $PQ = (-B/A)(QR)$.

Unfortunately, it's not so clear how to find QR . But we can find PR . Points P and R have the same x -coordinate, so PR equals the difference in the y -coordinates of P and R :

$$PR = y_0 - y_R = y_0 - \frac{-Ax_0 - C}{B} = \frac{Ax_0 + By_0 + C}{B}.$$



Now, we're close. We have PR , and we can relate PQ to QR . Finally, we use the Pythagorean Theorem to finish. We have $PR^2 = PQ^2 + QR^2$ and $QR = -(A/B)PQ$, so we have

$$\left(\frac{Ax_0 + By_0 + C}{B}\right)^2 = PQ^2 + \left(\frac{A^2}{B^2}\right)PQ^2.$$

Multiplying both sides by B^2 gives

$$(Ax_0 + By_0 + C)^2 = B^2 \cdot PQ^2 + A^2 \cdot PQ^2 = (A^2 + B^2)(PQ^2).$$

Dividing both sides by $A^2 + B^2$, then taking the square root of both sides, gives

$$PQ = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}.$$

We need the absolute value on the right side because length must be positive.

Our proof does not address the cases in which k is horizontal or vertical – you'll be asked to tackle these cases as an Exercise. Moreover, our proof assumes that the slope of k is positive, and that P is above the line. However, the proofs are essentially the same for other possible configurations in which k is neither horizontal nor vertical.

Also, our proof doesn't address the possibility that P is on k . We can quickly show that our formula works in this case. If $P = (x_0, y_0)$ is on the graph of $Ax + By + C = 0$, then we must have $Ax_0 + By_0 + C = 0$. When we substitute $Ax_0 + By_0 + C = 0$ in our formula, then our formula gives us a distance of 0, which is indeed the correct distance between point P and line k when P is on k . $\square$

Important: The distance between the point (x_0, y_0) and the graph of the equation $Ax + By + C = 0$ is



$$\frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}.$$

Let's check our formula by using it to solve Problem 17.18.

Important: Whenever you derive a formula, you should test it on a specific case.



Problem 17.20: Use your formula from Problem 17.19 to check your answer to Problem 17.18 by using the formula to find the distance between $(2, 7)$ and the graph of $x - 3y = 1$.

Solution for Problem 17.20: See if you can figure out where we go wrong here:

Bogus Solution: The distance between the point $(2, 7)$ and the graph of the equation $x - 3y = 1$ is



$$\frac{|2 - 3 \cdot 7 + 1|}{\sqrt{1^2 + 3^2}} = \frac{18}{\sqrt{10}} = \frac{18}{\sqrt{10}} \cdot \frac{\sqrt{10}}{\sqrt{10}} = \frac{9\sqrt{10}}{5}.$$

Our answer in the Bogus Solution doesn't match our answer to Problem 17.18. Is our formula wrong? No! We applied it incorrectly.

WARNING!! In our formula for the distance between a point and a line, the linear equation is in the form $Ax + By + C = 0$, not $Ax + By = C$.

So, we write our linear equation as $x - 3y - 1 = 0$. The distance between $(2, 7)$ and the graph of this equation is

$$\frac{|2 - 3 \cdot 7 - 1|}{\sqrt{1^2 + 3^2}} = \frac{20}{\sqrt{10}} = \frac{20}{\sqrt{10}} \cdot \frac{\sqrt{10}}{\sqrt{10}} = 2\sqrt{10},$$

which matches our original answer. $\square$

Our formula can also be used to solve less straightforward problems.

Problem 17.21: How many ordered pairs (x, y) satisfy both $14x - 48y + 49 = 0$ and $x^2 + y^2 = 1$?

Solution for Problem 17.21: We could solve the equation $14x - 48y + 49 = 0$ for x in terms of y and substitute the result into $x^2 + y^2 = 1$, but that will lead to some pretty ugly algebra. Before we do so, we think about analytic geometry, because the equation $x^2 + y^2 = 1$ is in the standard form of an equation whose graph is a circle. We like circles.

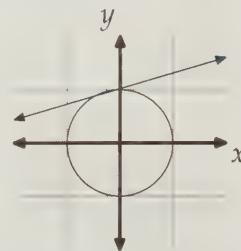
Concept: If an algebra problem has an equation whose graph is a circle, think about using analytic geometry.

The graph of the equation $x^2 + y^2 = 1$ is a circle with radius 1. The graph of the equation $14x - 48y + 49 = 0$ is a line, but does it intersect the circle? The graphs of both equations are shown at right. Unfortunately, we can't quite tell if the line hits the circle once or twice. And our graph might be off by just a little bit, so we can't even be sure if the line hits the circle at all!

Each point on the circle is 1 unit from the origin. So, if the line passes within 1 unit of the origin, it must intersect the circle twice, since it must go inside the circle. The distance between $(0, 0)$ and the graph of $14x - 48y + 49 = 0$ is

$$\frac{|14 \cdot 0 - 48 \cdot 0 + 49|}{\sqrt{14^2 + (-48)^2}} = \frac{49}{\sqrt{14^2 + (-48)^2}}.$$

Rather than multiplying out the squares in the denominator, we remember the Pythagorean triple $\{14, 48, 50\}$, so we know $14^2 + (-48)^2 = 50^2$. Therefore, the distance between the line and the origin is $49/50$. So, the line goes inside the circle, which means it intersects the circle twice. This tells us that there are two ordered pairs that satisfy both of the given equations. $\square$



Exercises

17.5.1 Find the distance between $(3, 4)$ and the graph of the equation $4x - 3y + 7 = 0$.

17.5.2 Find the coordinates of the points on the graph of $y = 10$ that are a distance of 5 from the graph of $x + 3y = 9$.

17.5.3 Our proof of the formula for the distance between a point and a line does not address the cases in which the line is either horizontal or vertical. Complete the proof of the formula by showing that it holds if the line is horizontal or vertical.

17.5.4 Find the distance between the graphs of $3x = 4y + 8$ and $3x = 4y + 17$.

17.5.5 At how many points does the graph of $2x - 3y = 48$ intersect the graph of $(x - 3)^2 + (y + 2)^2 = 100$?

17.6 Advanced Analytic Geometry Problems

Problems

Problem 17.22: In this problem, we find the area of the region bounded by the lines $2x + 3y = 21$ and $5x + 2y = 25$ and the coordinate axes.

- Graph the two equations. Do we have a simple formula to find the area of the region?
- Dissect the region into pieces you know how to handle, and use these pieces to solve the problem.

Problem 17.23:

- What point is the image of rotating the point $(-1, 3)$ an angle of 180° about $(-6, -7)$?
- What point is the image of rotating the point $(4, 5)$ an angle of 90° clockwise about the point $(-4, 2)$?
- What point is the image of reflecting the point $(5, -3)$ over the line $y = 8$?
- What point is the image of reflecting the point $(3, -1)$ over the graph of the line $2x - y + 5 = 0$?

Problem 17.24: Triangles ABC and ADE have areas 2007 and 7002, respectively, with $B = (0, 0)$, $C = (223, 0)$, $D = (680, 380)$, and $E = (689, 389)$. In this problem, we find the sum of all possible x -coordinates of A . (Source: AMC 12)

- What does the information about the area of $\triangle ABC$ tell us about the possible locations of point A ?
- What does the information about the area of $\triangle ADE$ tell us about the possible locations of point A ?
- Combine your observations from the first two parts. How many possible points A are there?
- What figure is formed when you connect all the possible points A you found in part (c)?
- Use your observation in part (d) to find the sum of all possible x -coordinates of A , *without actually finding any of the points that A could be*.

Extra! *Well done is better than well said.*

—Benjamin Franklin

Problem 17.25: Medians $\overline{XA}$ and $\overline{YB}$ of $\triangle XYZ$ are perpendicular. In $\triangle XYZ$, we have both $\angle YXZ = 90^\circ$ and $XY = 8$. In this problem we find YZ .

- Set the problem up on the Cartesian plane. Choose your origin wisely, then find the coordinates of X , Y , Z , A , and B . If you choose your origin wisely, you should only need one variable.
- Use the information in the problem to find the value of the variable.
- Find YZ .

Problem 17.26: A point P lies in the same plane as a given square of side 1. Let the vertices of the square, taken counterclockwise, be A , B , C , and D . Also, let the distances from P to A , B , and C , respectively, be u , v , and w . What is the greatest distance that P can be from D if $u^2 + v^2 = w^2$? (Source: AMC 12)

Problem 17.22: Find the area of the region bounded by the lines $2x + 3y = 21$ and $5x + 2y = 25$ and the coordinate axes.

Solution for Problem 17.22: We start by graphing the lines and shading the bounded region as shown below.

Important: Just as with most geometry problems, our first step with many analytic geometry problems is drawing a diagram.



$PQRS$ is a quadrilateral, but it's not one of our special types of quadrilateral. So, it's not immediately obvious how to find the area of the shaded region. Therefore, we use a tactic from our study of funky areas: we divide the shaded region into pieces we know how to handle. We draw diagonal $\overline{PR}$, which cuts the shaded region into two triangles.

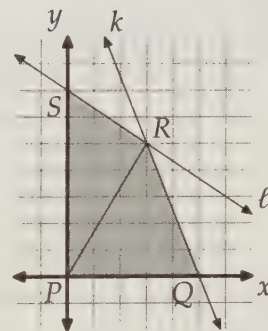
We could also have cut the region into two triangles by drawing $\overline{QS}$, but we can easily find the areas of $\triangle PQR$ and $\triangle PSR$. The graph of $5x + 2y = 25$ (line k in the diagram) intersects the x -axis at $(5, 0)$ and it intersects the graph of $2x + 3y = 21$ (line ℓ in the diagram) at $(3, 5)$. So, $PQ = 5$ and the altitude from R to $\overline{PQ}$ is 5. This gives us $[PQR] = (5)(5)/2 = 25/2$.

Similarly, the graph of $2x + 3y = 21$ meets the y -axis at $(0, 7)$, so $PS = 7$. The altitude from R to $\overline{PS}$ has length 3, so $[PSR] = (3)(7)/2 = 21/2$. Finally, we have

$$[PQRS] = [PQR] + [PSR] = \frac{25}{2} + \frac{21}{2} = 23.$$

We also could have solved this problem by dissecting $PQRS$ into a trapezoid and a triangle. See if you can solve the problem this way, as well. $\square$

Let's try some transformations on the Cartesian plane.



Problem 17.23:

- What point is the image of rotating the point $(-1, 3)$ an angle of 180° about $(-6, -7)$?
- What point is the image of rotating the point $(4, 5)$ an angle of 90° clockwise about the point $(-4, 2)$?
- What point is the image of reflecting the point $(5, -3)$ over the line $y = 8$?
- What point is the image of reflecting the point $(3, -1)$ over the line $2x - y + 5 = 0$?

Solution for Problem 17.23:

- Let X be $(-1, 3)$, let Y be $(-6, -7)$, and let Z be the image of X upon a 180° rotation about Y . Because the angle of rotation is 180° , we have $\angle XYZ = 180^\circ$. So, point Y is on $\overline{XZ}$. Furthermore, because Z is the image of X upon rotation about Y , we have $XY = YZ$. Since Y is also on $\overline{XZ}$, we know that Y is the midpoint of $\overline{XZ}$. So, if Z is (a, b) , we have

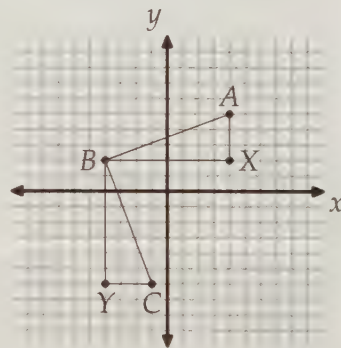
$$\left(\frac{a + (-1)}{2}, \frac{b + 3}{2} \right) = (-6, -7).$$

From this, we find $a = -11$ and $b = -17$, so the image of X upon 180° rotation about Y is $(-11, -17)$.

We also could have solved this problem by noting that because Y is the midpoint of $\overline{XZ}$, we know that the translation that maps X to Y also maps Y to Z . To get from X to Y , we go left 5 units and down 10 units. So, to get from Y to Z , we also go left 5 units and down 10 units, to $(-6 - 5, -7 - 10) = (-11, -17)$.

- Let $(4, 5)$ be point A and $(-4, 2)$ be point B as shown in the diagram at right. If point C is the image of A upon a 90° clockwise rotation about B , then we must have $AB = CB$ and $\overline{AB} \perp \overline{CB}$. The slope of $\overline{AB}$ is $3/8$, so we know that the slope of $\overline{CB}$ is $-8/3$. This means that C is on the line through B with slope $-8/3$. But where on this line is it?

We can find point C by noting that to get from A to B , we go down 3 units and left 8 units. So, to get from point B to C , we go right 3 units and down 8 units, which means C is $(-4 + 3, 2 - 8) = (-1, -6)$. This is illustrated in the figure at right. We first build right triangle $\triangle ABX$ so that $\overline{AX}$ is vertical and $\overline{BX}$ is horizontal. When we rotate this triangle 90° clockwise, we get $\triangle BYC$, in which $BY = BX = 8$ and $CY = AX = 3$. Therefore, point C is 8 units below and 3 units to the right of B .



- The point $(5, -3)$ is 11 units below the horizontal line $y = 8$, so its image is 11 units directly above $y = 8$. Therefore, the image of $(5, -3)$ upon reflection over the graph of $y = 8$ is $(5, 8 + 11) = (5, 19)$. We can quickly check this by noting that the graph of $y = 8$ is the perpendicular bisector of the segment with endpoints $(5, -3)$ and $(5, 19)$. This tells us that $(5, 19)$ is the image of $(5, -3)$ upon reflection over the graph of $y = 8$.
- Let line k be the graph of $2x - y + 5 = 0$. Let P be $(3, -1)$, and let R be its image upon reflection over k . Because R is the image of P upon reflection over k , $\overline{PR}$ and k are perpendicular. The slope of k is 2, so the slope of $\overline{PR}$ is $-1/2$. Therefore, R is on the line through P with slope $-1/2$. A point-slope form of this line's equation is

$$y - (-1) = -\frac{1}{2}(x - 3).$$

But which point on the graph of this line is R ? If we can find the point where $\overline{PR}$ meets k , we can use this point to find R , because this intersection point is the midpoint of $\overline{PR}$. Let the intersection point of $\overline{PR}$ and k be Q . We have the equations of both $\overline{PR}$ and k , so we can find Q by solving the system of equations

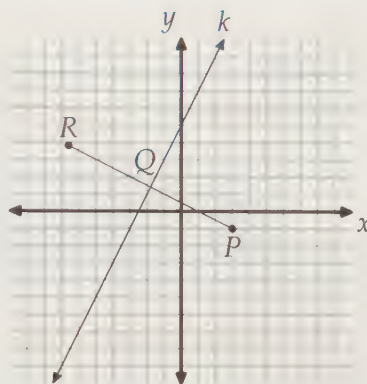
$$\begin{aligned} 2x - y + 5 &= 0, \\ y + 1 &= -\frac{1}{2}(x - 3). \end{aligned}$$

Adding these two equations gives $2x + 6 = -\frac{x}{2} + \frac{3}{2}$. Solving this equation gives $x = -9/5$. Substituting this into either of the above equations gives $y = 7/5$. So, point Q is $(-9/5, 7/5)$.

Now we're ready to find R . Let the coordinates of R be (x_R, y_R) . Because Q is the midpoint of $\overline{PR}$, we must have

$$\left(\frac{3 + x_R}{2}, \frac{-1 + y_R}{2}\right) = \left(-\frac{9}{5}, \frac{7}{5}\right).$$

Solving $(3 + x_R)/2 = -9/5$ and $(-1 + y_R)/2 = 7/5$ gives us $(x_R, y_R) = (-33/5, 19/5)$. We could also have found the coordinates of R by noting that to get from P to Q , we go $24/5$ units to the left and $12/5$ units up. So, to get from Q to R , we go units $24/5$ to the left and $12/5$ units up from $(-9/5, 7/5)$ to $(-33/5, 19/5)$.



□

Problem 17.24: Triangles ABC and ADE have areas 2007 and 7002, respectively, with $B = (0, 0)$, $C = (223, 0)$, $D = (680, 380)$, and $E = (689, 389)$. Find the sum of all possible x -coordinates of A . (Source: AMC 12)

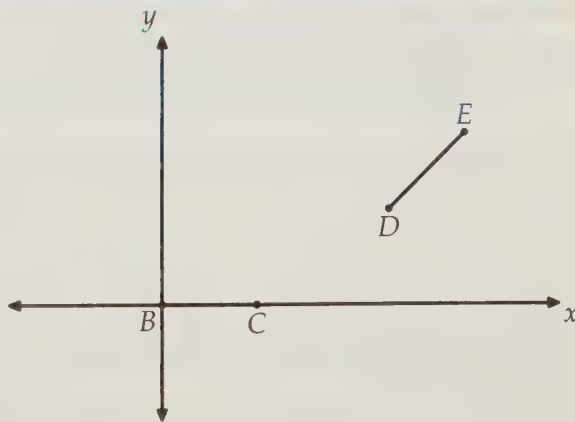
Solution for Problem 17.24: The huge numbers in the problem make it hard to draw our graph to scale, so we start with a rough sketch. We'll have to use the information about triangles ABC and ADE to determine the possible points A . We start with the fact that $[ABC] = 2007$. Base $\overline{BC}$ of this triangle has length 223, so if we let h_1 be the length of the altitude to side $\overline{BC}$ of $\triangle ABC$, we have

$$h_1 = \frac{2[ABC]}{BC} = \frac{2(2007)}{223} = 18.$$

This tells us that A is 18 units from $\overleftrightarrow{BC}$. Since $\overline{BC}$ is along the x -axis, we now know that A is on the graph of either $y = 18$ or $y = -18$.

Triangle ABC gave us some information about A ; let's take a look at $\triangle ADE$. The distance formula tells us that $DE = 9\sqrt{2}$, so if we let h_2 be the length of the altitude from A to $\overline{DE}$, we have

$$h_2 = \frac{2[ADE]}{DE} = \frac{2(7002)}{9\sqrt{2}} = \frac{2(778)}{\sqrt{2}} = 778\sqrt{2}.$$



So, point A is $778\sqrt{2}$ away from $\overleftrightarrow{DE}$. We could find the equations of the two lines that are $778\sqrt{2}$ away from $\overleftrightarrow{DE}$, but that looks like it might be pretty difficult. Instead, we go back to our diagram and see if we can find anything to simplify the problem.

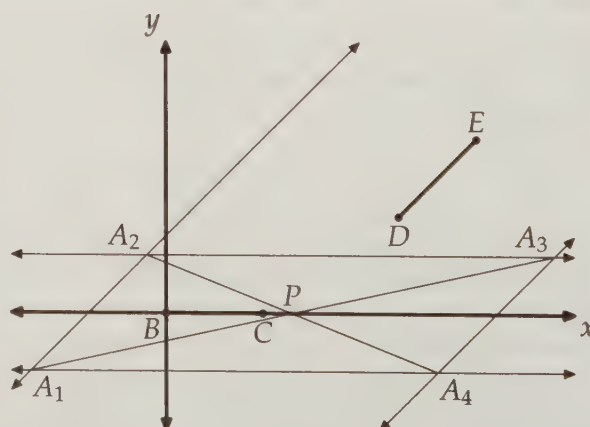
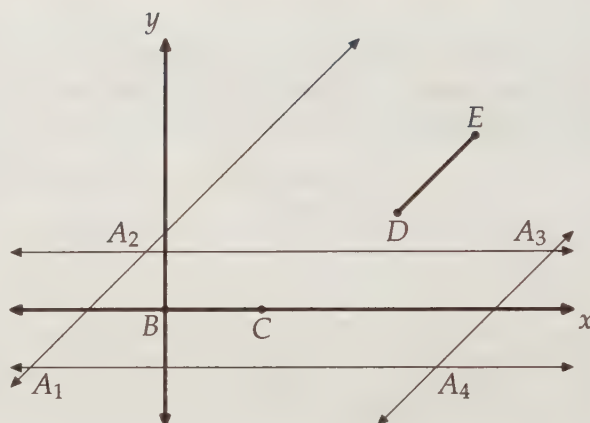
Concept: One reason we draw diagrams for analytic geometry problems is that they might inspire geometric shortcuts.

We've found that A must be on one of two lines parallel to $\overleftrightarrow{BC}$, and that these two lines are equidistant from $\overleftrightarrow{BC}$. Similarly, we found that A must be on one of two lines parallel to and equidistant from $\overleftrightarrow{DE}$. We add all four of these lines to our diagram. The four possible points A are where pairs of these four lines intersect. We label these A_1 , A_2 , A_3 , and A_4 , as shown. $A_1A_2A_3A_4$ is a parallelogram because the opposite sides of $A_1A_2A_3A_4$ are along parallel lines.

We could find the coordinates of the vertices of the parallelogram $A_1A_2A_3A_4$, but that looks like a lot of work. Instead, we think a little bit to see if we can avoid all that work. Because $A_1A_2A_3A_4$ is a parallelogram, we think about what's special about parallelograms, and how to use these special properties to find the sum of the x -coordinates of the vertices.

Specifically, we focus on parallelogram facts that might be used to discover something about this sum of x -coordinates. This gets us thinking about midpoints, since finding the midpoint of a segment requires adding coordinates. This, in turn, leads us to the fact that the diagonals of a parallelogram bisect each other, which means they have the same midpoint. In other words, if x_1 , x_2 , x_3 , and x_4 are the x -coordinates of A_1 , A_2 , A_3 , and A_4 , respectively, then the x -coordinate of the intersection of diagonals $\overline{A_1A_3}$ and $\overline{A_2A_4}$ equals both $(x_1 + x_3)/2$ and $(x_2 + x_4)/2$. We want $x_1 + x_2 + x_3 + x_4$, so we know we're close!

We let P be the intersection of diagonals $\overline{A_1A_3}$ and $\overline{A_2A_4}$, and let the coordinates of P be (x_P, y_P) . We just saw above that $x_P = (x_1 + x_3)/2 = (x_2 + x_4)/2$. So, we have $x_1 + x_3 = x_2 + x_4 = 2x_P$, which means that our desired sum $x_1 + x_2 + x_3 + x_4$ equals $4x_P$. All we have to do now is find P . We know P is the intersection of diagonals $\overline{A_1A_3}$ and $\overline{A_2A_4}$, but we would like to find a faster way to find P than finding all the vertices of $A_1A_2A_3A_4$. Our diagram gives us our slick approach. Because P is the midpoint of both diagonals, it is equidistant from opposite sides $\overline{A_1A_2}$ and $\overline{A_3A_4}$, and it is equidistant from opposite sides $\overline{A_2A_3}$ and $\overline{A_1A_4}$. Aha! $\overleftrightarrow{DE}$ is also equidistant from $\overline{A_1A_2}$ and $\overline{A_3A_4}$, and $\overleftrightarrow{BC}$ is equidistant from $\overline{A_2A_3}$ and $\overline{A_1A_4}$. So, P is the intersection point of $\overleftrightarrow{BC}$ and $\overleftrightarrow{DE}$. This point is easy to find!



Because B and C are on the x -axis, $\overrightarrow{BC}$ is the x -axis. So, all points on $\overrightarrow{BC}$ satisfy $y = 0$. This means the y -coordinate of P is 0. The slope of $\overrightarrow{DE}$ is $(389 - 380)/(689 - 680) = 1$, and the line passes through $(680, 380)$, so $\overrightarrow{DE}$ is the graph of the equation $y - 380 = 1(x - 680)$. Rearranging this equation gives $x - y = 300$. Point P is on the graph of this line, and the y -coordinate of P is 0, so the x -coordinate of P is 300. Therefore, our desired sum of x -coordinates is $4(300) = 1200$. $\square$

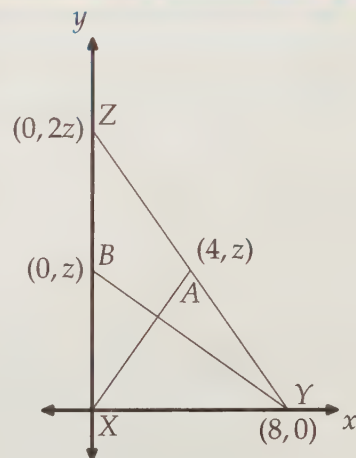
We've seen how geometric insights can help us with analytic geometry problems. Let's take a look at a couple examples of how we can use analytic geometry to tackle challenging geometric problems. Just as we saw when using analytic geometry for geometric proofs, one of the main keys in each problem is conveniently setting up the problem on the Cartesian plane.

Problem 17.25: Medians $\overline{XA}$ and $\overline{YB}$ of $\triangle XYZ$ are perpendicular. If $\angle YXZ = 90^\circ$ and $XY = 8$, find YZ .

Solution for Problem 17.25: We have two big clues to try analytic geometry: midpoints and perpendicular lines. We let X , the vertex of the right angle, be the origin. This allows us to place Y on the x -axis and Z on the y -axis. Because $XY = 8$, we can let Y be $(8, 0)$. Then, we can let Z be $(0, 2z)$ for some value of z . We use $2z$ instead of just z because we know we'll be working with midpoints in this problem. If we find z , we can use the distance formula to find YZ , since

$$YZ = \sqrt{(2z - 0)^2 + (0 - 8)^2} = \sqrt{4z^2 + 64} = 2\sqrt{z^2 + 16}.$$

We now can find the coordinates of A and B in terms of z . Because A is the midpoint of $\overline{YZ}$, its coordinates are $(4, z)$. Because B is the midpoint of $\overline{XZ}$, its coordinates are $(0, z)$. But now what?



Concept: When you're stuck on a problem, focus on information you haven't used yet.

We haven't used the fact that $\overline{XA} \perp \overline{YB}$. Because $\overline{XA} \perp \overline{YB}$, the product of the slopes of $\overline{XA}$ and $\overline{YB}$ is -1 . The slope of $\overline{XA}$ is $z/4$ and the slope of $\overline{YB}$ is $-z/8$. Therefore, we have

$$\left(\frac{z}{4}\right)\left(\frac{-z}{8}\right) = -1.$$

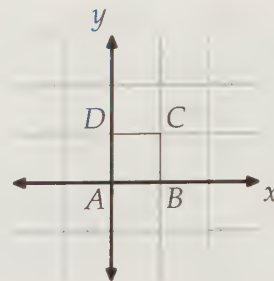
This gives us $-z^2/32 = -1$, so $z^2 = 32$. Now we can find YZ using our formula from above. We have

$$YZ = 2\sqrt{z^2 + 16} = 2\sqrt{32 + 16} = 2\sqrt{48} = 8\sqrt{3}.$$

$\square$

Problem 17.26: A point P lies in the same plane as a given square of side 1. Let the vertices of the square, taken counterclockwise, be A, B, C , and D . Also, let the distances from P to A, B , and C , respectively, be u, v , and w . What is the greatest distance that P can be from D if $u^2 + v^2 = w^2$? (Source: AMC 12)

Solution for Problem 17.26: Squares are easy to represent on the Cartesian plane. We even know each side has length 1, so we can let the vertices of our square be $A = (0, 0)$, $B = (1, 0)$, $C = (1, 1)$, and $D = (0, 1)$.



Our given equation $u^2 + v^2 = w^2$ involves squares of distances. This is another clue that analytic geometry might be helpful: squaring the distances gets rid of the square root sign in the distance formula. We let the coordinates of P be (x, y) , so we can use the distance formula to find:

$$\begin{aligned} u^2 &= PA^2 = (x - 0)^2 + (y - 0)^2, \\ v^2 &= PB^2 = (x - 1)^2 + (y - 0)^2, \\ w^2 &= PC^2 = (x - 1)^2 + (y - 1)^2. \end{aligned}$$

Substituting these into $u^2 + v^2 = w^2$ gives us

$$x^2 + y^2 + (x - 1)^2 + y^2 = (x - 1)^2 + (y - 1)^2.$$

The $(x - 1)^2$ terms cancel, and a little rearranging gives

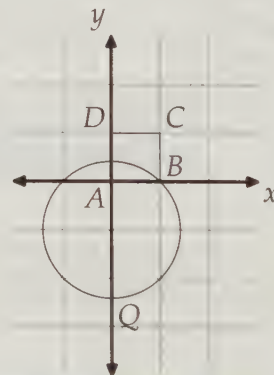
$$x^2 + y^2 + 2y - 1 = 0.$$

The graph of this equation is a circle. To learn more about this circle, we complete the square in y on the left. This gives us

$$x^2 + (y + 1)^2 = 2.$$

This means that (x, y) is on a circle with center $(0, -1)$ and radius $\sqrt{2}$. Moreover, we can reverse our steps above to see that any point on the circle satisfies the restrictions placed on point P in the problem. So, now our problem is to find the point on this circle that is farthest from point D .

We add this circle to our diagram at right. Because both the center of the circle and point D are on the y -axis, the point on the circle that is the farthest from D is on the opposite side of the circle from D . In other words, it is the 'bottommost' point of the circle, which we label as Q in the diagram. Because D is 2 units from the center of the circle, and the radius of the circle is $\sqrt{2}$, the distance between D and Q is $2 + \sqrt{2}$, which is therefore our desired greatest possible distance. $\square$



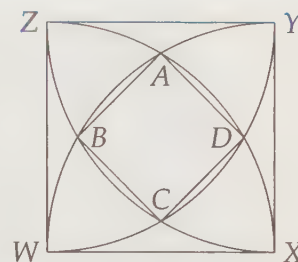
Exercises

17.6.1 What is the image of $(5, 6)$ under each of the following transformations:

- | | |
|---|--|
| (a) A 180° rotation about the origin. | (d) A 90° counterclockwise rotation about $(-3, 2)$. |
| (b) The translation that maps $(2, -3)$ to $(0, 6)$. | (e) Reflection over the graph of $y = x$. |
| (c) Reflection over the graph of $x = -3$. | (f) Reflection over the graph of $2x + 3y = -5$. |

17.6.2 Two nonadjacent vertices of a rectangle are $(4, 3)$ and $(-4, -3)$, and the coordinates of the other two vertices are integers. How many different such rectangles are there? (Source: AMC 12)

17.6.3 WXYZ is a square with $WX = 6$. Quarter circles centered at each vertex of WXYZ are drawn inside the square as shown. The intersections of these quarter circles form square ABCD shown.



- (a) Find the area of ABCD with analytic geometry.
- (b) Find the area of ABCD without analytic geometry.

17.6.4 Two parallel lines intersect the x -axis at points that are 5 apart. The same two lines intersect the y -axis at points that are 12 apart. How far apart are the two lines?

17.6.5 Let point C be $(3, -2)$. Points A and B are on the graph of $2x - y = -7$ such that $\triangle ABC$ is equilateral. Find AB.

17.6.6 Solve Problem 17.25 without using analytic geometry. **Hints:** 42

17.6.7★ Given that $x^2 + y^2 = 14x + 6y + 6$, what is the largest possible value that $3x + 4y$ can have? (Source: AHSME) **Hints:** 2

17.6.8★ Let $\overline{AC}$ and $\overline{BD}$ be two perpendicular chords of a circle with radius 8, and let the two chords intersect at P. Find all possible values of $PA^2 + PB^2 + PC^2 + PD^2$.

17.7 Summary

Important:



The graph of the equation $y - y_1 = m(x - x_1)$ is a line through (x_1, y_1) with slope m . This is called a **point-slope form** of the equation.

The **standard form** of a linear equation is $Ax + By = C$, where, if possible, A, B, and C are integers, A is positive, and A, B, and C have no common factors besides 1.

The **slope-intercept form** of a linear equation is $y = mx + b$, where m is the slope of the line and b is the y -coordinate of the y -intercept.

Important:



The distance in the plane between the points (x_1, y_1) and (x_2, y_2) is

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

This is often referred to as the **distance formula**.

Important:



The midpoint of the segment with endpoints (x_1, y_1) and (x_2, y_2) is

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right).$$

Important:

If two non-vertical lines have the same slope, then they are parallel. Conversely, if two non-vertical lines are parallel, then they have the same slope.

If two non-vertical lines are perpendicular, then the product of their slopes is -1 . Conversely, if the product of the slopes of two lines is -1 , then the two lines are perpendicular.

Important:

The **standard form** of an equation whose graph is a circle is

$$(x - h)^2 + (y - k)^2 = r^2,$$

where h, k , and r are constants with $r > 0$. The center of the circle is (h, k) and the radius of the circle is r .

Problem Solving Strategies

Concepts:

- When the algebra in an analytic geometry problem starts to get ugly, try using geometric facts to simplify the problem.
- When given the coordinates of the vertices of a triangle, check if the triangle is special in any way. Most notably, check if the triangle is a right triangle. If it is, this fact will probably simplify the problem, because we know so much about right triangles.
- Interpreting an equation as a geometric figure on the Cartesian plane can often help us solve problems.
- When setting a geometry problem up on the Cartesian plane, choose the origin and the axes in a convenient matter. Often this means letting the origin be the vertex of a right angle in the problem, so that the sides of the angle are along the axes.
- Solving a problem in two ways is a good way to check your answer.
- Checking intermediate steps while working on problems will also help you catch a lot of errors.
- When we set up a geometry problem involving midpoints on the Cartesian plane, we often use $2a, 2b, 2c$, etc., for coordinates rather than just a, b, c , etc. This helps us avoid expressions involving fractions when we find midpoints of segments in the problem.

Continued on the next page. . .

Concepts: . . . continued from the previous page



- Algebra is not the only tool we have to solve problems about the Cartesian plane. Combining geometric insights with algebra can lead to very nice solutions.
- When stuck on a problem, focus on information that you haven't used yet.
- If an algebra problem has an equation whose graph is a circle, think about using analytic geometry.
- One reason we draw diagrams for analytic geometry problems is that they might inspire geometric shortcuts.

REVIEW PROBLEMS

- 17.27** Find an equation, in standard form, whose graph is the line through $(-5, 2)$ that is perpendicular to the graph of $5x - 2y + 7 = 0$.
- 17.28** Find the area of the triangle with vertices $(3, 4)$, $(8, 4)$, and $(-6, -6)$.
- 17.29** Let A be $(-2, 4)$, B be $(-7, 1)$, and C be $(-1, -5)$.
- Find an equation whose graph is a line that contains the median from A to the midpoint of $\overline{BC}$.
 - Find the length of the median from A to $\overline{BC}$.
 - Find the length of the altitude from A to $\overline{BC}$.
 - Find the area of $\triangle ABC$.
- 17.30** Find the constant a such that the graph of $(2a - 3)x + (3a - 1)y = 3$ is parallel to the segment with endpoints $(2, -4)$ and $(-1, 2)$.
- 17.31** Graph the equation $x^2 - 6x + y^2 + 4y = -3$. Find the area enclosed by your graph.
- 17.32** Let P be $(-2, 5)$ and Q be $(8, -3)$. Find the standard form of the equation of the circle with $\overline{PQ}$ as a diameter.
- 17.33** Let point X be $(10, 1)$, and let C be the graph of $(x - 1)^2 + (y + 2)^2 = 10$.
- Let Y be the point on C that is closest to X . Find XY .
 - ★ Find the coordinates of Y .
- 17.34** Find all possible values of a such that the distance between the point $(a, 3)$ and the graph of $5x - 12y = 8$ is 7.
- 17.35** Point A is $(-3, 5)$. The midpoint of $\overline{AB}$ is $(4, 2)$ and the midpoint of $\overline{AC}$ is $(3, 1)$. What is BC ?

17.36 Point O is the origin and point B is $(0, 13)$. Point A has positive coordinates such that $AO = 5$ and $AB = 12$. Find the coordinates of A .

17.37 What is the image of $(5, -2)$ when reflected over each of the following:

- (a) The x -axis.
- (b) The graph of $x = -3$.
- (c)★ The graph of $2x - y = -1$.

17.38 Find the length of the segment that is a chord of both the graph of $x^2 + y^2 = 36$ and the graph of $x^2 + 2x + y^2 = 30$.

17.39 Use analytic geometry to show that if $AB = AC$ in $\triangle ABC$, then the altitude from A to $\overline{BC}$ is also the median from A to $\overline{BC}$.

17.40 Two adjacent vertices of a square are $(-4, 5)$ and $(-3, 7)$. What are all possible points besides these two that could be vertices of this square?

17.41 Show that the image of the point (a, b) upon reflection over the line $y = x$ is (b, a) .

17.42 Point X is $(2, 3)$ and point Y is $(6, 3)$. Find all possible points P such that $\triangle XYP$ is equilateral.

17.43 Use analytic geometry to show that the sum of the squares of the lengths of the diagonals of a parallelogram equals the sum of the squares of the lengths of the sides of the parallelogram.

17.44 Find the equation whose graph is the circle that passes through the points $(-1, 5)$, $(4, 4)$, and $(5, 9)$.

17.45 In triangle ABC , let D, E, F be the midpoints of $\overline{BC}$, $\overline{AC}$, $\overline{AB}$, respectively.

- (a) Show that triangles ABC and DEF have the same centroid.
- (b) Let G_1, G_2, G_3 be the centroids of triangles AFE, BDF, CED , respectively. Show that triangles ABC and $G_1G_2G_3$ have the same centroid.

17.46 Points X and Y are on $\overline{AB}$ such that $AX = XY = BY$. If A is $(0, 9)$ and B is $(4, 0)$, what is the slope of the line through X and the origin?

17.47 Point P is $(6, 2)$. Points Q and R are on the x -axis such that $\angle PQR = 30^\circ$ and $\angle PRQ = 60^\circ$. What is the area of $\triangle PQR$?

17.48 Point X is $(4, -3)$, point Y is $(-2, 5)$, and point Z is $(c, 3)$.

- (a) For what values of c is $\triangle XYZ$ isosceles?
- (b) For what values of c is $\triangle XYZ$ a right triangle?

Challenge Problems

17.49 Find an equation whose graph is a circle that passes through the origin and both the x -intercept and the y -intercept of the graph of $x - 3y = 18$.

17.50

- (a) The graph of the equation $3x - y = 0$ is rotated 90° clockwise about the origin to produce line k . Find an equation whose graph is line k .
- (b) The graph of the equation $3x - y = 6$ is rotated 90° clockwise about the origin to produce line k . Find an equation whose graph is line k .

17.51 The graphs of $x^2 + y^2 = 4 + 12x + 6y$ and $x^2 + y^2 = k + 4x + 12y$ intersect when k satisfies $a \leq k \leq b$, and for no other values of k . Find $b - a$. (Source: AMC 12)

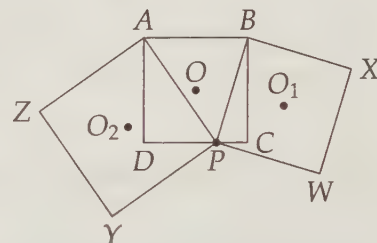
17.52 How many points are equidistant from the x -axis, the y -axis, and the graph of the equation $x + y = 2$? (Source: AMC 12)

17.53 Point X is on diameter $\overline{PQ}$ of a circle. Prove that if $\overline{AB}$ is a chord of the circle parallel to $\overline{PQ}$, then $XA^2 + XB^2 = XP^2 + XQ^2$.

17.54 Let U be the point $(-4, 3)$ and V be the point $(2, -1)$.

- (a) Let A be $(-1, 3)$. Let A' be the image when A is rotated 90° clockwise about U . Let A'' be the image when A' is rotated 90° clockwise about V . Find A'' .
- (b) Let B be $(7, -2)$. Let B' be the image when B is rotated 90° clockwise about U . Let B'' be the image when B' is rotated 90° clockwise about V . Find B'' .
- (c) Find the midpoint of $\overline{AA''}$ and the midpoint of $\overline{BB''}$. Notice anything interesting?
- (d)★ Let C be (a, b) . Let C' be the image when C is rotated 90° clockwise about U . Let C'' be the image when C' is rotated 90° clockwise about V . Show that your observation in part (c) can be used to quickly find C'' for any a and b .

17.55 Let $ABCD$ be a square, and let P be a point on side $\overline{CD}$. Construct squares $BPWX$ and $APYZ$ externally on $\overline{BP}$ and $\overline{AP}$, respectively, as shown. Let O , O_1 , and O_2 be the centers of squares $ABCD$, $BPWX$, and $APYZ$, respectively. Show that quadrilateral OO_1PO_2 is a parallelogram.



17.56★ Find the largest value of y/x for pairs of real numbers (x, y) that satisfy $(x - 3)^2 + (y - 3)^2 = 6$. (Source: AHSME)

17.57★ Let $\overline{A_1A_2}$ and $\overline{B_1B_2}$ be two perpendicular chords of a circle with radius r , each passing through a fixed point P . Show that $(A_1A_2)^2 + (B_1B_2)^2$ is independent of the position of the chords. (In other words, show that for all pairs of perpendicular chords of this circle that meet at P , the sum of the squares of the lengths of the chords is the same.)

Extra! ...fain would I turn back the clock and devote to French or some other language the hours I spent upon algebra, geometry, and trigonometry, of which not one principle remains with me. Stay! There is one theorem painfully drummed into my head which seems to have inhabited some corner of my brain since that early time: "The square on the hypotenuse of a right-angled triangle is equal to the sum of the squares on the other two sides!" There it sticks, but what of it, ye gods, what of it?

—Jessie B. Rittenhouse